Two barriers to equitable access: WCAG conformance and pre-consent tracking on Ecuadorian and top-ranked university websites

Gleiston Guerrero-Ulloa[a], Andrea Zúñiga Paredes[b] and Efraín Díaz-Macías[a]

[a]Facultad de Ciencias de la Computación, Universidad Técnica Estatal de Quevedo, Quevedo, Ecuador; [b]Ministerio de Educación del Ecuador, Quito, Ecuador

ARTICLE HISTORY
Compiled November 11, 2026

ABSTRACT
Applicants find programmes, apply and enrol through the institutional website, so what the site does to its visitors decides who can use it on equal terms. We audited the home pages of the 63 Ecuadorian higher education institutions and of 63 top-ranked universities in August 2026 for two such conditions, conformance with the Web Content Accessibility Guidelines (WCAG) and tracking before any consent is given. Ecuadorian institutions gave visitors less to act on: 38.1% identified a privacy contact, against 85.7%, and 45% of their failing accessibility nodes were at the basic level A, against 19%. Pre-consent tracking did not differ between the groups, but it did differ with the visitor's location: the same pages set tracking cookies for 67.8% of visits from Ecuador and 60.3% of visits from Germany. A second campaign the next day showed the same direction, though that contrast was not significant, and its network records locate the decision. At 13 of the 14 sites where Microsoft set cookies for a visitor in Ecuador, a visitor in Germany still sent requests to Microsoft's servers and received none, so the vendor, not the university, decided. Protection therefore depends on where users are and on third parties they cannot see.

KEYWORDS
web accessibility; online privacy; cookie consent; third-party tracking; digital inequality; university websites

1. Introduction

The institutional website is the channel through which a prospective student finds programme information, submits an application, completes enrolment and reaches library, tutoring and administrative services. A barrier embedded in that channel is therefore not a cosmetic defect but a restriction on access to the institution. Universal access research has long framed this as a design problem rather than a remediation problem: environments and services should be usable by the widest possible range of people from the outset, and accessibility standards are the operational expression of that principle for digital products (Hermosa-Ramírez, 2025). The Web Content Accessibility Guidelines (WCAG), currently in version 2.2 (World Wide Web Consortium, 2024), provide the testable criteria, and several legal regimes make them binding, among them the European Accessibility Act (European Parliament and Council of

CONTACT Gleiston Guerrero-Ulloa. Email: gguerrero@uteq.edu.ec

the European Union, 2019) and, in the United States, the Americans with Disabilities Act (ADA) and Section 508 of the Rehabilitation Act (United States Congress, 1986, 1990).

Personal data protection is a second barrier operating on the same channel and, we argue, on the same principle. A visitor who must accept unexplained tracking in order to consult a study programme, or who cannot find out who processes their application data or how to object, faces a cost of participation that is unevenly distributed: it falls hardest on users with least legal literacy, least technical means to refuse and least ability to seek an alternative provider. Read this way, opaque privacy governance and inaccessible interfaces are two mechanisms of the same phenomenon, namely the unequal cost of using a public-facing service. This framing motivates studying them together rather than as two separate compliance checklists.

Both barriers also act through what users can see and do. Research in this journal has shown that users' privacy literacy and self-efficacy depend in part on how easily they can reach privacy information (Meier & Krämer, 2025), that self-efficacy, alongside concern, explains whether users act to protect their privacy (Epstein & Quinn, 2025), and that the way a consent request or a privacy policy is presented changes how attentive users are to what they disclose and how far they trust the provider (Lindegren et al., 2021; Magrizos et al., 2026). On the accessibility side, the developers who build such pages have been found to know the guidelines poorly and to find them hard to engage with (Cao & Loiacono, 2022; Spyridonis & Daylamani-Zad, 2021). Users in different countries also weigh privacy differently (Li et al., 2022). A university page that names no privacy contact, sets tracking cookies before the visitor can respond, or labels its links so that a screen reader cannot announce them therefore shapes the behaviour of the people who use it, and does so before they can exercise any choice.

Empirical work has treated the two literatures separately. On accessibility, audits of university portals report widespread barriers in the highest-ranked universities assessed against WCAG 2.2 (Krittika & Shimray, 2025), in European institutions (Krejtz et al., 2025), in the Gulf region (Fakrudeen, 2025) and in public online services generally (Aljedaani, 2025; Iniesto & Rodrigo, 2025), with a long-documented lag in Latin America (Acosta-Vargas, González, & Luján-Mora, 2020; Campoverde-Molina et al., 2023). On privacy, large-scale measurement has established that written rules and observed behaviour advance at different speeds: the General Data Protection Regulation (GDPR) (European Parliament and Council of the European Union, 2016) multiplied notices and banners without proportionate change in tracking (Degeling et al., 2019; Kretschmer et al., 2021; Trevisan et al., 2019), consent interfaces frequently fail the legal conditions for valid consent (Gray et al., 2021; Habib et al., 2022; Matte et al., 2020; Nouwens et al., 2020), cookie managers often do not block what they claim to block (Bollinger et al., 2022; Sanchez-Rola et al., 2019), and in higher education specifically, policies have been observed to change to look compliant rather than to be compliant (Munot et al., 2025). To our knowledge, no study has measured both barriers on the same set of institutional sites, with the same protocol and on the same date.

Ecuador is an informative setting for that measurement. Its Organic Law on the Protection of Personal Data (Ley Orgánica de Protección de Datos Personales, LOPDP) entered into force in 2021 and was implemented by regulation only in 2023 (Asamblea Nacional del Ecuador, 2021; Presidencia de la República del Ecuador, 2023), placing it in the most recent wave of a regional diffusion process that comparative work has shown to be uneven in commitment and enforcement (Carrillo & Jackson, 2022; Chavarría-Mora, 2025). The country also has a closed, enumerable population of 63

recognised higher education institutions (HEIs), which allows a complete census rather than a sample.

This article addresses five research questions.

- **RQ1.** How do Ecuadorian HEIs and a top-ranked international benchmark group differ in the privacy governance visible on their institutional home page, namely a published privacy notice, a cited legal framework, an enumeration of data-subject rights and an identified privacy contact?
- **RQ2.** How do the two groups differ in observed pre-consent behaviour, that is, in the cookies set before any interaction with a consent interface, when both are observed from the same vantage point outside the European Union?
- **RQ3.** How do the two groups differ in automatically detectable WCAG conformance, and does the difference lie in the volume of failures or in the conformance level at which they occur?
- **RQ4.** Within each group, is measured accessibility associated with observed restraint in pre-consent tracking?
- **RQ5.** Does the pre-consent behaviour of the same page depend on where the *visitor* is located, and if so, is the difference decided by the institution or by the third parties whose code its page embeds?

The contributions are the following. First, a paired measurement of accessibility and pre-consent tracking on 126 university home pages with a single, fully specified and reproducible protocol, released with its configuration, its tracking-cookie taxonomy and its raw output. Second, an inferential reanalysis that separates the differences the design can support from those it cannot, including an explicit equivalence test and a power statement for the null results the study reports. Third, a four-vantage replication of the live audit, from Ecuador, Germany, the United Kingdom and the United States, together with a five-vantage campaign that records the network requests sent to each vendor, which converts a legal conjecture into a measurement: the protection a visitor receives is shown to depend on the visitor's location and, for the largest vendor involved, to be decided by the vendor rather than by the institution. Fourth, an indicator-by-jurisdiction mapping that separates observed conduct from legal non-compliance, since most of the twelve jurisdictions studied impose no general prior-consent duty for cookies.

The remainder of the article is organised as follows. Section 2 reviews the two literatures and states the gap. Section 3 describes the design, the sampling, the codebook, the audit protocol and the statistical analysis. Section 4 reports the results for each research question. Section 5 interprets them, derives implications for users, practice and policy, and states the threats to validity. Section 6 concludes and sets out the future work the present design cannot deliver.

2. Related work

2.1. Web accessibility as a condition of equitable access

Universal design treats accessibility as a property to be built into a service rather than added to it. The tradition is not, however, terminologically settled, and the lack of settlement has methodological consequences: Persson et al. (2015) trace the parallel histories of universal design, inclusive design, accessible design and design for all and show that the absence of a shared definition weakens both measurability and confor-

mance assessment, which is why the present article fixes its operational definitions before reporting any result. Two commitments of that tradition bear directly on this study. The first is that a barrier is a property of the encounter between a person and a service rather than of the person, so an audit describes what a service does to the range of people obliged to use it. The second is distributive: Abascal et al. (2016) argue that universal accessibility formulated without regard to socioeconomic and geopolitical context reaches only a fraction of the population it claims to serve, and that exclusion by location must be addressed alongside exclusion by ability. That argument is the one Section 5.2 returns to, because the boundary this study measures turns out to be geographical rather than functional. On the instrument side, Etxabe-Antia et al. (2025) review 596 accessibility recommendations across digital technologies and find that established web technologies converge on WCAG while emerging ones do not, which supports the choice of WCAG as the reference standard for institutional web pages; and media accessibility research has examined how the same principle interacts with ecological and context-of-use considerations (Hermosa-Ramírez, 2025). Applied to higher education, the systematic review by Campoverde-Molina et al. (2023) covers university website accessibility worldwide and reports persistent non-conformance across regions, methods and time periods. More recent single-study evidence is consistent with that picture: Krittika and Shimray (2025) evaluate the top 25 universities of the Quacquarelli Symonds (QS) world ranking and the top 25 of its Indian ranking against WCAG 2.2 and find failures in both sets; Krejtz et al. (2025) report on the accessibility information published by European universities; and Fakrudeen (2025) reach similar conclusions for the Gulf region. Outside higher education, Aljedaani (2025) and Iniesto and Rodrigo (2025) document the same dominant error types on public-sector portals, principally insufficient contrast and links without an accessible name. In Latin America, Acosta-Vargas, González, and Luján-Mora (2020) and Acosta-Vargas, Ramos-Galarza, Salvador-Ullauri, Chanchi, and Jadán-Guerrero (2020) provide comparative evidence of a regional lag, and Maldonado-Garcés et al. (2025) extend the argument from the digital to the physical campus in Ecuador.

Two methodological cautions follow from this literature and shape the present design. First, automated tools demonstrate non-conformance but cannot certify conformance: Fischer et al. (2025) quantify the coverage of automated checkers and show that they address only a minority of the WCAG success criteria, so an automated pass is not evidence of an accessible site. Second, results depend on the tool, on the rule set enabled and on the rendering conditions, which is why this study reports its full configuration rather than only its tool name.

2.2. Privacy governance and web privacy measurement

The measurement literature on web privacy provides both the instruments and the cautionary findings for the present study. Libert (2015) and Englehardt and Narayanan (2016) established large-scale browser-based measurement of third-party tracking and showed that the phenomenon is concentrated, persistent and largely invisible to users. After the GDPR became applicable, Degeling et al. (2019) and Trevisan et al. (2019) found that notices and banners proliferated without a proportionate reduction in tracking, and Kretschmer et al. (2021) reached comparable conclusions over a longer window. A second strand shows that the interfaces themselves are frequently non-compliant: Matte et al. (2020) found that a large share of banners built on the Interactive Advertising Bureau (IAB) Europe Transparency and Consent Framework

registered a consent signal inconsistent with the user's choice; Nouwens et al. (2020), Gray et al. (2021) and Habib et al. (2022) document design patterns that make acceptance the path of least resistance; and Sanchez-Rola et al. (2019) and Bollinger et al. (2022) show that consent-management platforms often fail to block the trackers they purport to gate. On the textual side, privacy policies remain long and hard to read (Belcheva et al., 2023; Fabian et al., 2017).

A third strand, directly relevant here, treats the location of the observer as an experimental variable rather than as a fixed property of the study. Dabrowski et al. (2019) crawled the same sites from clients in the European Union and in the United States and found that 26.6% of the cookie-setting sites placed persistent cookies for the United States client and none for the European one, an asymmetry that was almost absent in the opposite direction. van Eijk et al. (2019) crawled from eighteen vantage points and reached a more conservative conclusion, that the top-level domain of the site explains more of the variation in cookie notices than the location of the visitor does, with `.com` sites the exception. Rasaii et al. (2023) confirm the pattern on the notice side, reporting consent banners 56% more prevalent when a site is visited from within the European Union. The three studies agree that where a measurement is taken from changes what is measured; they do not agree on what governs the difference.

In higher education, Liu and Khalil (2023) review privacy issues in learning analytics and Mutimukwe et al. (2026) examine online proctoring, both concerned with internal systems rather than the public portal. Closest to the present work, Munot et al. (2025) analyse how universities reshaped their privacy policies after the GDPR and distinguish compliance achieved by design from compliance achieved by appearance.

2.3. The gap addressed here

Three gaps follow from the two literatures reviewed above, and the design of this study is a response to each of them.

The first is a gap of scope. The two literatures share a subject, the institutional website, and an underlying concern, the cost that a service imposes on the people who must use it, but they have not been brought together on a common set of sites. Accessibility studies do not measure tracking; privacy measurement studies do not audit conformance. Because the two are always measured apart, a question that matters for practice cannot be answered at all: whether an institution that commits to one form of access also commits to the other, or whether the two are independent properties that a single institutional commitment does not deliver. Section 4.6 tests that association directly, which requires both barriers to be measured on the same sites, on the same day, with one protocol.

The second is a gap of vantage point. The geo-differential literature is built almost entirely on an axis between the European Union and the United States, and it attributes what it finds to the site: a site either discriminates by visitor location or it does not. Two consequences follow. Latin America appears in none of these designs, so what a visitor located under a young data-protection regime actually receives has not been measured; and because the unit of attribution is the site, the mechanism is left unresolved, since an asymmetry observed at the site cannot distinguish a decision taken by the operator of the site from one taken by the third parties whose code the site embeds. This distinction is not academic: it determines whether a remedy addressed to universities can work at all. Section 4.7 measures from four vantage points, one of them Ecuadorian, attributes the difference to named vendors, and uses the net-

work records of a second, five-vantage campaign to establish whether each difference is decided by the vendor or before the request reaches it.

The third is a gap of population. Studies of university websites, in both literatures, rest overwhelmingly on ranked or convenience samples, which by construction describe institutions selected for visibility. A complete national population supports a different claim, because it carries no sampling error and no selection on prominence, and because it is the population that a national regulator is actually obliged to supervise. Setting such a population against an elite benchmark under an identical protocol is what allows the differences reported in Section 4 to be read as differences between the two groups rather than as differences between two studies. That combination, two barriers, four vantage points and a complete national population against a benchmark, is what this article contributes.

3. Methods

3.1. Design and unit of analysis

The study is a cross-sectional, two-group comparison carried out in August 2026. The unit of analysis is the *institutional home page and the first-level privacy notice linked from it*. This is a deliberately narrow unit and it is stated explicitly because it determines what the results can mean: a document that exists on a faculty subdomain but is not reachable from the home page is coded as absent, since a prospective applicant arriving at the institutional portal would not find it. The one-click allowance governs where the notice may be found, not how far its content may extend. The notice itself is coded as present when it is reachable from the home page or one click below it, whereas the framework, rights and privacy-contact indicators are coded from the content of that notice alone. Material stated a further click below the notice is therefore recorded as absent even where it exists, and Section 3.4 reports the two institutions in which this occurred.

The comparison is a benchmarking exercise against an aspirational ceiling, not an estimate of a country effect. The benchmark group is not a sample of "the rest of the world": it is the extreme upper tail of an international ranking distribution, and it differs from the Ecuadorian population in budget, technical staffing and regulatory maturity as well as in country. Every difference reported in Section 4 is therefore open to interpretation as an effect of institutional resources rather than of jurisdiction, and no causal attribution to country is made anywhere in this article.

3.2. Sampling

The Ecuadorian group is a complete census of the 63 universities and polytechnic institutions recognised by the national higher-education authorities, the Secretariat of Higher Education, Science, Technology and Innovation (Secretaría de Educación Superior, Ciencia, Tecnología e Innovación, SENESCYT) and the Council of Higher Education (Consejo de Educación Superior, CES), as listed on 11 August 2026. Institutions created in 2024 or later are included and marked in Appendix D; there is one, USECIPOL, and readers can exclude it if they consider the comparison unfair to an institution that has not yet completed its first academic cycle.

The benchmark group was constructed by consensus of three rankings in their most recent editions: the QS World University Rankings 2026, the Times Higher Education

(THE) World University Rankings 2026 and the Academic Ranking of World Universities (ARWU) 2025 (QS Quacquarelli Symonds, 2025; ShanghaiRanking Consultancy, 2025; Times Higher Education, 2025). From each ranking the top 75 was taken; each institution received the mean of its three positions, with absence from a ranking's top 75 imputed as position 76; the number of rankings in which an institution appeared took precedence over that mean, so the 46 institutions present in all three head the list and the remaining 17 were drawn, by mean position, from those present in two. The cut-off institution was the University of Queensland and the highest-placed institution excluded was the London School of Economics. The group size was set at 63 to match the Ecuadorian census. Figure E1 in Appendix E shows the resulting distribution by country; the United States contributes 24 institutions, the United Kingdom 7, China 6 and Australia 5, with the remainder spread over eleven further jurisdictions.

The aggregation rule involves three arbitrary choices, namely the imputation constant, the depth of the ranking cut and the lexicographic priority given to ranking coverage. These were fixed before the selection was made and were not varied in the analysis reported here; Section 5.6 records the dependence of the composition on them as a limitation rather than claiming it as robust.

3.3. Dimensions, operational definitions and codebook

Six documentary dimensions were coded from the institution's own published material, and a further set of indicators was measured directly: transport security by connection test, and pre-consent cookies, tracking cookies, consent banner, consent-management platform and WCAG conformance by the automated audit. The distinction is between what an institution *states* and what its site *does*, and it is maintained throughout the article. The operational definitions are given in full in Appendix A; the essential points are the following.

Privacy notice. Coded as present when a document whose stated purpose is to describe the processing of personal data is reachable from the home page. Coded **P** (partial) when the document exists but its stated scope covers only part of the institution's processing, for example admissions only, postgraduate programmes only or a mobile application only. Coded NV (not verifiable) when the site could not be retrieved or the document was not machine-readable.

Applicable legal framework cited. Coded as present when the notice names a specific legal instrument. Generic formulations such as "applicable law" or "data protection legislation" were coded as absent, and a named instrument that does not apply to the institution was coded **P** (partial). In Ecuador the relevant instrument is the LOPDP and its implementing regulation (Asamblea Nacional del Ecuador, 2021; Presidencia de la República del Ecuador, 2023); elsewhere the framework applicable to the institution was accepted, as set out in the indicator-by-jurisdiction mapping of Appendix B.

Data-subject rights listed. Coded as present when the notice enumerates at least three of the rights of access, rectification, erasure, restriction, objection and portability, together with a route to exercise them. A generic mention of "your rights" without enumeration was coded as absent, and an enumeration given without any route to exercise the rights was coded **P** (partial).

Privacy contact or data protection officer. Coded as present when the notice gives a named role or a dedicated address or form for privacy matters. A general institutional contact form was coded as absent. We distinguish throughout between the *controller*, which is the institution itself and is therefore never absent, and the *data protection*

officer (DPO) or privacy contact, which is the designated route through which a data subject can act.

Dedicated cookie policy. Coded as present when a document specifically about cookies or similar tracking technologies is published and reachable under the same rule as the privacy notice. A section on cookies inside a general privacy notice was coded as absent, since the indicator records whether the institution treats tracking as a subject requiring its own statement. This is a documentary indicator and is distinct from the live banner and CMP measurements of Section 3.5: a published policy is what the institution says, a banner is what the site does. It was coded for the benchmark group only, because the Ecuadorian matrix records no counterpart, and it is therefore reported descriptively in Section 4.1 rather than compared.

Accessibility statement declared. Coded as present when an accessibility statement, an accessibility policy or an accessibility toolbar is published and reachable under the same rule as the privacy notice. Accessibility mentioned only inside another document was coded as absent. Like the cookie policy this indicator records a declaration, not a property of the interface; the measured conformance of that interface is reported separately in Section 4.4.

Measured dimensions. Transport security was recorded by connection test: a site was coded as present when its home page is served over HTTPS with a certificate chain that a current browser accepts without warning. It is reported under the heading of transport security and not of data protection, because it measures confidentiality in transit and says nothing about what the site does with the data once received. It was tested separately from the automated audit, since the audit browser was configured to ignore certificate errors in order to reach sites with faulty chains. Pre-consent cookies, tracking cookies, consent banner and consent-management platform (CMP) were recorded by the automated audit described in Section 3.5, as was WCAG conformance.

Treatment of the partial category. Three documentary indicators admit a partial code, which is assigned 24 times across the two groups: 18 in the benchmark group and 6 in the Ecuadorian census. A twenty-fifth partial code was recorded at one Ecuadorian institution, UPSE, on the accessibility statement, an indicator for which the codebook provides no partial category; it is kept in the released matrix as recorded and, like every partial cell, counted as not meeting the indicator. Every proportion reported in this article counts a partial cell as not meeting the indicator. The reason is substantive rather than conservative: a notice whose scope covers only part of the institution's processing, an instrument named that does not govern the institution, and an enumeration of rights with no route to exercise them each leave the data subject in the position the indicator is meant to detect. Because the choice is consequential for one indicator in particular, its effect is quantified in Section 4.1 rather than asserted, and the per-cell codes are released so that any other aggregation can be computed.

3.4. Documentary coding and its verification

The documentary matrix was first compiled on 11 August 2026 with the assistance of a generative artificial intelligence system, which retrieved each institution's home page and first-level privacy notice as text and proposed a code for each indicator; the declaration at the end of this article records this use. The retrieval did not execute JavaScript, so content rendered dynamically could not be read, which is one of the reasons a cell could be recorded as not verifiable. The three authors and one external collaborator then took part in coding and reviewing the matrix, but their individual

codings were not retained, so the study cannot report an inter-coder reliability coefficient for the documentary indicators. This is a material limitation and is stated as such in Section 5.6.

To obtain a bounded indication of coding error, a targeted verification sub-study was carried out on 14 August 2026. Six institutions of the benchmark group, selected purposively as those whose individual coding carried the greatest weight in the analysis, were re-examined manually, cell by cell, against the live site. The four governance indicators were covered for each of them, namely the first-level privacy notice, the specific legal instrument named, the enumeration of data-subject rights and the privacy contact or data protection officer, giving 24 cells in all. The dedicated cookie policy and the accessibility statement were not part of the sub-study. Two discrepancies were found. One institution had been coded as publishing no privacy notice when a notice is in fact linked from the home-page footer, and the cell was corrected to present. One institution had been coded as citing an applicable framework when its notice names no instrument and uses only generic formulations, and the cell was corrected to absent. Both corrections are incorporated in every figure and table of this article, and the two cells are marked in Appendix C. The sub-study also revealed a systematic effect of the depth-one unit of analysis: at two further institutions, rights and framework are stated one click below the first-level notice and are therefore recorded as absent under our definition even though the content exists. This is a property of the measure, not an error, but it means that the governance indicators reported here should be read as "visible at the first level" rather than as "published anywhere by the institution".

Because the six institutions were selected purposively rather than at random, two errors in 24 cells is not a reliability estimate and is neither an upper nor a lower bound on the error rate of the full matrix; it is reported only to show that the rate is not negligible. Independent double coding by human coders, without assistance, of at least 30% of the sample, with a Cohen's κ per indicator and reconciliation of every discrepancy, is the first item of the future work set out in Section 6. The resulting asymmetry, between the reliability evidence available for the documentary indicators and that available for the live ones, is stated as a limitation in Section 5.6.

3.5. Automated audit protocol

Each of the 126 sites was visited once, on 14 August 2026, from a residential connection in Ecuador, with a read-only script. The full configuration is given here so that the measurement can be repeated.

- *Browser and driver.* Chromium driven by Playwright 1.62.1 (Microsoft, 2026) under Node.js v24.18.1, launched headless with the flags `--no-sandbox` and `--disable-dev-shm-usage`. A fresh browser context was created per site, so no state carried over between sites.
- *Viewport and user agent.* Playwright's default desktop viewport of 1280×720 CSS pixels was used, with the user-agent string of Chrome 128 on Windows 10 (64-bit). The viewport is reported because it determines the outcome of the contrast, reflow and target-size criteria.
- *Navigation.* `page.goto` with `waitUntil: 'domcontentloaded'`, a 45 s timeout and one retry on failure, followed by a fixed 4 s wait to allow deferred scripts and consent platforms to load. The context was created with `ignoreHTTPSErrors: true` so that sites with certificate problems could still

be audited.[1] No scrolling and no interaction with any banner were performed at any point.

- *Cookie state.* The full cookie jar of the context was read before any interaction. A cookie was classified as *tracking* when its name matched a declared allow-list of analytics, advertising and session-replay families. Two versions of that list exist and the distinction matters for reading the figures, so both are reproduced in full in Appendix A, where the patterns belonging only to the second are marked. The list applied at collection covered Google Analytics and Google Ads, the YouTube and DoubleClick families, Meta, TikTok, Microsoft Clarity and UET, X, Hotjar, Siteimprove, Matomo, PixelYourSite, Pinterest, Adobe, Tealium, Baidu, Quantcast and LiveIntent. It was extended after collection, once the observed cookie names had been inspected; the families added at that point, the reason for adding them and the effect of the extension on every reported figure are given in Section 3.7. *Every figure in this article is computed with the extended list*, obtained by reclassifying the stored cookie names, so no measurement was repeated. Cookies not matching the list, including first-party session and language cookies, were counted in the total but not as tracking.
- *Consent platform and banner.* A CMP was recorded when a signature of one of thirteen named platforms matched the `src` of a script, iframe or link element, when one of a fixed white-list of CMP global objects was defined, or when `window.__tcfapi` was a function. A banner was recorded when a fixed or sticky element of non-zero height below 600 px contained text matching cookie, consent or privacy. Both are heuristics and their false-negative rate was not measured.
- *Accessibility.* axe-core 4.13 (Deque Systems, 2025) was injected into the loaded page and run over the whole document with `resultTypes` set to violations, passes and incomplete. The rule set was selected by tag: `wcag2a`, `wcag2aa`, `wcag21a`, `wcag21aa`, `wcag22a` and `wcag22aa`, plus `wcag2aaa` and `wcag21aaa`. The reference standard is therefore WCAG 2.2 at levels A and AA (World Wide Web Consortium, 2024) together with the level-AAA rules of WCAG 2.0 and 2.1 (World Wide Web Consortium, 2008, 2018); no WCAG 2.2 level-AAA rules were enabled, because axe-core 4.13 does not implement them.
- *Derived measures.* Each violation was assigned its success criterion, its principle and its conformance level from the axe tags, with the highest level taken when a rule carries several. A site was recorded as *reaching level A* when it produced no level-A violation, and as having *no automatic failure* when it produced no violation at any enabled level. The number of checks axe flags for manual review was recorded per site but is not analysed here.

Two properties of this protocol matter for interpretation and are not incidental. First, enabling the level-AAA rules is a substantive choice: the dominant AAA rule is enhanced contrast (success criterion 1.4.6, requiring a 7:1 ratio), and any statement about the AAA share of failures is conditional on that rule being active. Section 4.4 therefore reports the level distribution both with and without the AAA rules. Second, the single vantage point in Ecuador is what the pre-consent measurement means: what is observed is what an Ecuadorian visitor experiences, not what a visitor located in the European Union would experience.

[1] This is why the five Ecuadorian sites recorded as lacking a valid certificate chain in Section 4.3 nonetheless appear in every accessibility and cookie figure: the audit browser reached them, and only the separate connection test of Section 3.3 records the deficiency.

3.6. Manual validation of three success criteria

Automated auditing detects a subset of conformance failures, and for three of the criteria that matter most to the argument of this article the automated result was verified by hand on a subsample of 15 sites, seven from the benchmark group and eight from the Ecuadorian census. The criteria are 1.1.1 non-text content, 1.4.3 contrast (minimum) and 2.4.4 link purpose (in context), all of WCAG 2.2 (World Wide Web Consortium, 2024), which is also published as an international standard (International Organization for Standardization, 2025).

The instrument is anchored in the Accessibility Conformance Testing (ACT) Rules of the W3C Web Accessibility Initiative, whose format became a W3C Recommendation in February 2026 (Fiers et al., 2026). This anchoring is a methodological decision and not a formality. ACT Rules are written to produce reproducible outcomes across evaluators and tools, and their outcome vocabulary is *applicable*, *passed*, *failed* and *inapplicable*: it admits no intermediate category. The unit of coding is therefore the individual element rather than the site, and what counts as an applicable element is fixed by the rule rather than decided by the evaluator. Appendix A gives the rule mapping (W3C Web Accessibility Initiative, 2026) and the resulting operational definitions.

Two consequences of that choice are worth stating. First, exclusions that would otherwise be a judgement of the evaluator follow from the rule: rule `afw4f7` applies to any visible character in a text node, so visually hidden text, skip links and elements rendered at zero font size are outside its scope by definition rather than by discretion. Second, the notion of a link's context is the *programmatically determined link context* of WCAG, which is a closed list, and not whichever ancestor a DOM traversal happens to reach; a generic wrapping element does not supply context.

The coding is exhaustive rather than sampled: every element the rule declares applicable is coded, with no cap on the number of elements and no ordering by suspicion. It is also blind to the automated audit. The browser-console collectors `act_recode.js` and `act_images.js` neither load nor query axe-core, and no axe-core output was consulted while coding. This property is what makes the comparison in Section 4.5 informative: an instrument primed by the automated tool's own flags could only confirm what that tool had already suspected, whereas this one had no access to them.

Two independent human codings were carried out. The first was collected on 8 September 2026 and the second on 9 September by a second evaluator working from a written protocol, without access to the first coding or to the draft of this article. On the three rules that the rule itself decides, `23a2a8`, `afw4f7` and `c487ae`, the two codings agree on 3 186 of 3 189 element-level comparisons, that is 99.9 %, with a Cohen's κ of 0.996 (95 % CI 0.991 to 1.000) at an identical prevalence of the majority category, 86.7 % in both; agreement on which elements the rules apply to is 99.1 % ($\kappa = 0.981$). That figure estimates the reproducibility of the instrument across independent operators. The remaining rules require a human judgement, and the second evaluator resolved on 11 September 2026 every judgement row that could change a site verdict. Under SC 2.4.4 both codings rest on human judgement: on the 62 comparisons available, all of them the links of a single institution, observed agreement is 82.3 % with a Cohen's κ of 0.178 (95 % CI -0.262 to 0.619), an interval that includes zero, so the estimate neither establishes nor refutes reliability and no inference is drawn from it. Under SC 1.1.1 the judgement rows of the first coding, whether an image's accessible name serves a purpose equivalent to the image, were resolved with the assistance of a generative artificial intelligence system, as the declaration at the end of

this article records. So that every verdict reported here rests on human judgement, the site verdicts of Section 4.5 are those of the second coding. They coincide with the first coding in 43 of the 45 site-by-criterion cells; the two exceptions are SC 1.1.1 at ECOTEC and at UC Berkeley, which the first coding fails on assisted judgements, and Section 4.5 reports them as a sensitivity reading.

The subsample of fifteen sites was selected purposively and the selection criterion was not recorded. It is therefore not a probability sample of either group, and the counts reported in Section 4.5 describe the sites examined, without extrapolation to the census.

3.7. Multi-vantage replication

Because the audit described above observed from one location only, it could not distinguish two possibilities with opposite legal readings: that sites treat every visitor alike, or that they present different interfaces depending on where the visitor is. To separate them, the live audit was replicated from four vantage points on 15 August 2026, with every other parameter unchanged.

The Ecuadorian vantage point was a residential connection in Cuenca (ETAPA EP, autonomous system AS27668) with no virtual private network (VPN). The other three were reached through a commercial VPN: Frankfurt, Germany (DigitalOcean AS14061 for two passes, Limestone Networks AS46475 for one); London, United Kingdom (Zenlayer AS21859); and Miami, United States (Limestone Networks AS46475). Three passes were run per vantage point, twelve in total, and each site's outcome was consolidated across passes by majority rule. The public Internet Protocol (IP) address and its geolocation were verified at the start of every pass, after every 25 sites and at the end, giving seven checks per pass; all 84 checks matched the declared vantage point, and the recordings are released with the data. One further pass was aborted automatically by that mechanism when the VPN had not yet switched country, and was discarded.

Two design properties of this replication matter. First, the Ecuadorian and United States vantage points are both outside the European Economic Area but differ in network type, residential against data centre, whereas the German, British and United States endpoints are all data-centre addresses; the German-versus-United States contrast is therefore free of any residential-versus-data-centre artefact. Second, two quantities were tracked as controls: the number of WCAG violations per site, which cannot depend on consent logic and serves as the rendering control, and the volume of cookies that are not attributable to any advertising vendor. A third property strengthens the contrast further and was not designed for: one of the three German passes was routed through Limestone Networks AS46475, the same autonomous system as the United States vantage point. Where that pass agrees with the other two German passes and differs from the United States ones, the difference cannot be an artefact of the hosting network, since the network is the same.

The tracking-cookie taxonomy was extended for this analysis after inspection of the cookie names observed. The families added at that point are the LinkedIn Insight Tag, TapAd, StackAdapt, Snapchat, ContentSquare, Sourcebuster and Google's Privacy Sandbox cookie, each added in full, together with further patterns belonging to families the original list already covered in part: four of the YouTube and DoubleClick family, five of the Microsoft advertising family, two of X, and one of TikTok. Appendix A marks every pattern belonging only to the extended list. The extended list, given

in full in Appendix A, is applied by reclassifying the stored cookie names, so no remeasurement was required and every reported figure derives from the same recorded observations. The extended list is used throughout this article, including in the group comparison of Section 4.2, so that one taxonomy governs every reported figure. Its effect on that comparison was checked rather than assumed, and it is small but not null: reclassification moves one Ecuadorian site across the tracking threshold, because its only matching cookies belong to the Sourcebuster family added at this point, and leaves every other site in both groups unchanged. The Ecuadorian count therefore rises from 40 to 41 of 63 while the benchmark count stays at 43 of 63, and Table 1, Figure 2 and the intervals, tests and equivalence bounds of Section 4.2 are computed on that basis. The extension matters considerably more at the level of individual cookies, which is what Section 4.7 attributes to vendors.

3.8. Network records

The cookie jar records whether a vendor's cookie is present but not why it is absent. An absent cookie can mean that the vendor's code ran and did not set it, or that the request to the vendor was never sent because the site, its consent platform, its tag manager or another third party withheld it. To separate the two, a second campaign was run on 16 August 2026 with version 3.0 of the same instrument, which keeps every measurement parameter of the four-vantage replication unchanged and adds a passive record of the requests the page sends to each vendor family, identified by domain, and of whether any response to them carried a `Set-Cookie` header. The instrument does not set the browser's time zone or language, which therefore follow the operating system of the measuring computer; every pass of both campaigns ran on the same platform (Windows, x64) with the same software versions and user-agent string, and the non-Ecuadorian vantage points were reached through a VPN, which changes the network address but not the browser's settings. It was run from five vantage points with three passes each: the residential connection in Cuenca used on 15 August, and datacentre endpoints in Frankfurt, London, Miami and Zurich. The Frankfurt and Miami endpoints were on the same autonomous system, Limestone Networks AS46475, in all six of their passes, so the contrast between Germany and the United States holds the hosting network constant. Location was verified seven times per pass, and all 105 checks matched the declared vantage point; a five-site pilot pass from Ecuador preceded the campaign and is not analysed. Outcomes were consolidated by majority over the three passes, and 121 sites returned valid passes at all five vantage points. For a site at which a vendor family sets a cookie for a visitor in Ecuador, the absence of that cookie at another vantage point is attributed to the vendor when the request to the vendor's domains was still sent, and to a layer upstream of the vendor when it was not. The analysis is reproduced by `vendor_layer.py` in the deposited code.

3.9. Statistical analysis

Proportions are reported with 95% Wilson score intervals. Group differences are reported as risk differences with 95% Newcombe intervals and as odds ratios with 95% Woolf intervals, using the Haldane–Anscombe correction of 0.5 where a cell is empty. Significance is assessed with the two-sided Fisher exact test, which is appropriate throughout given the expected cell counts in the regional breakdown. Because twelve indicators are compared, p values are adjusted within that family by the Holm–

Bonferroni procedure and the adjusted values are reported.

For the indicator on which no difference was found, an equivalence test was performed rather than an argument from a non-significant p value. Two one-sided tests (TOST) were run at margins of 10, 15 and 20 percentage points. The margins were not pre-specified, which is stated openly, and the result is reported as a bound on what the design can establish rather than as a demonstration of equivalence.

Vantage points are compared with the exact McNemar test, which is the appropriate test when the same site is observed under two conditions, and with Cochran's Q for the four-way comparison; discordant-pair counts and exact p values are reported alongside every such comparison. Multiplicity is handled in three separate families, each declared before the analysis it governs: the twelve indicators of the group comparison, the six pairwise comparisons between the four vantage points of 15 August, and the ten pairwise comparisons between the five vantage points of 16 August. The Holm–Bonferroni procedure is applied within each family, the adjusted value governs every claim of difference made in this article, and no claim is made across families. The association between accessibility and pre-consent tracking is analysed within each group separately as well as pooled, because pooling two populations with different marginal distributions on both variables invites Simpson's paradox. Percentages of failing nodes by conformance level are rounded to the nearest integer, which is why the benchmark distribution sums to 101 rather than 100. All analyses were carried out in Python 3.10.3 with NumPy 2.2.6, SciPy 1.15.3 and Matplotlib 3.10.9; the analysis scripts are released with the data.

3.10. Ethical considerations

The study involves no human participants and no personal data: the observations are properties of public web pages. It nonetheless carries reputational risk for the institutions observed, and three measures were adopted in response. First, no institution is named in the body of the article for its documentary indicators or for the group comparison of pre-consent behaviour; those results appear institution by institution only in the appendices and in the deposit, where the coding date, the definition and the limitations of the measure are stated alongside them. The body names institutions in three places only: the fifteen sites of the manual validation in Section 4.5 and Table 4, the sites at which a named vendor's cookies were observed in Section 4.7, and the sites that could not be measured at every vantage point, named in the same section. In all three the site is the unit of the observation, and readers need it to check that observation against the deposited records. Second, the cells of the six institutions whose individual coding carried most weight in the analysis were manually re-verified, and the two corrections found are reported in Section 3.4. Third, the results are framed as a description of observed conduct on one date, not as a determination of legal non-compliance, since, as Appendix B sets out, most of the jurisdictions studied impose no general duty of prior consent for cookies.

A fourth consideration applies to the commercial third parties named in Section 4.7. They are named because their observed conduct is the finding of that section, not as an allegation: what is reported is the cookie state recorded on public pages on a single date, the number of sites is given alongside every name, and no inference is drawn about any vendor's internal configuration, intent or legal position. The observations, the classification rules and the per-site output are released with the data, so that any vendor named can reproduce or contest the measurement.

The script observed a conservative load profile, one request sequence per site with no crawling beyond the home page, and did not attempt to circumvent access controls.

4. Results

Table 1 reports every indicator for both groups with the risk difference, the odds ratio and the Holm-adjusted p value. Figure 1 displays the risk differences with their Newcombe intervals, grouped by domain, and Figure 2 gives the Wilson intervals of the pre-consent proportions. Appendices C and D give the institution-level matrices.

Table 1. Indicator-by-indicator comparison of the benchmark group and the Ecuadorian census ($n = 63$ per group, August 2026). RD: risk difference in percentage points, benchmark minus Ecuador, with 95% Newcombe interval. OR: odds ratio with 95% Woolf interval. p_{H}: two-sided Fisher exact test, Holm-adjusted over the twelve comparisons.

Indicator	Benchmark	Ecuador	RD [95% CI]	OR [95% CI]	p_{H}
Privacy governance (documentary)					
Privacy notice published	59/63 (93.7)	39/63 (61.9)	+31.7 [+17.6, +44.7]	9.1 [2.9, 28.2]	<0.001
Framework cited	47/63 (74.6)	31/63 (49.2)	+25.4 [+8.4, +40.4]	3.0 [1.4, 6.4]	0.045
Rights listed	47/63 (74.6)	35/63 (55.6)	+19.0 [+2.4, +34.3]	2.4 [1.1, 5.0]	0.235
Privacy contact or DPO	54/63 (85.7)	24/63 (38.1)	+47.6 [+31.3, +60.4]	9.8 [4.1, 23.3]	<0.001
Pre-consent behaviour (live audit)					
Tracking cookies set	43/63 (68.3)	41/63 (65.1)	+3.2 [−13.0, +19.1]	1.2 [0.5, 2.4]	1.000
At least one cookie set	56/63 (88.9)	55/63 (87.3)	+1.6 [−10.2, +13.4]	1.2 [0.4, 3.4]	1.000
Consent banner shown	22/63 (34.9)	15/63 (23.8)	+11.1 [−4.8, +26.3]	1.7 [0.8, 3.7]	0.961
CMP detected	12/63 (19.0)	14/63 (22.2)	−3.2 [−17.2, +11.0]	0.8 [0.3, 2.0]	1.000
Accessibility					
Accessibility statement	47/63 (74.6)	7/63 (11.1)	+63.5 [+47.9, +74.2]	23.5 [8.9, 61.9]	<0.001
No level-A failure	25/63 (39.7)	5/63 (7.9)	+31.7 [+17.2, +44.9]	7.6 [2.7, 21.7]	<0.001
No failure at any level	6/63 (9.5)	0/63 (0.0)	+9.5 [+1.8, +19.3]	14.4 [0.8, 260]	0.193
Transport security					
Valid TLS certificate	63/63 (100.0)	58/63 (92.1)	+7.9 [+0.6, +17.3]	11.9 [0.7, 221]	0.288

Note: percentages in parentheses. "No level-A failure" denotes the absence of automatically detectable level-A violations and is not a claim of WCAG conformance, which automated tools cannot establish (Fischer et al., 2025). Two cells of the benchmark group were corrected after manual re-verification, as described in Section 3.4. The tracking indicator is computed under the extended cookie taxonomy of Appendix A.

4.1. Privacy governance (RQ1)

The governance differences were the largest and the most robust in the study. A privacy notice was reachable from the home page at 59 of the 63 benchmark sites (93.7%) and at 39 of the 63 Ecuadorian sites (61.9%), a difference of 31.7 percentage points (odds ratio 9.08, adjusted $p < 0.001$). The requirement that the notice name a specific legal instrument reduced both figures, to 74.6% and 49.2% respectively (odds ratio 3.03, adjusted $p = 0.045$).

The widest gap was in the route by which a data subject can act. A privacy contact or data protection officer was identified at 85.7% of benchmark sites and at 38.1% of Ecuadorian sites, a difference of 47.6 percentage points (odds ratio 9.75, adjusted $p < 0.001$). The enumeration of data-subject rights differed in the same direction, 74.6% compared with 55.6%, but this difference did not survive correction for multiple comparisons (adjusted $p = 0.235$) and is reported here as suggestive only.

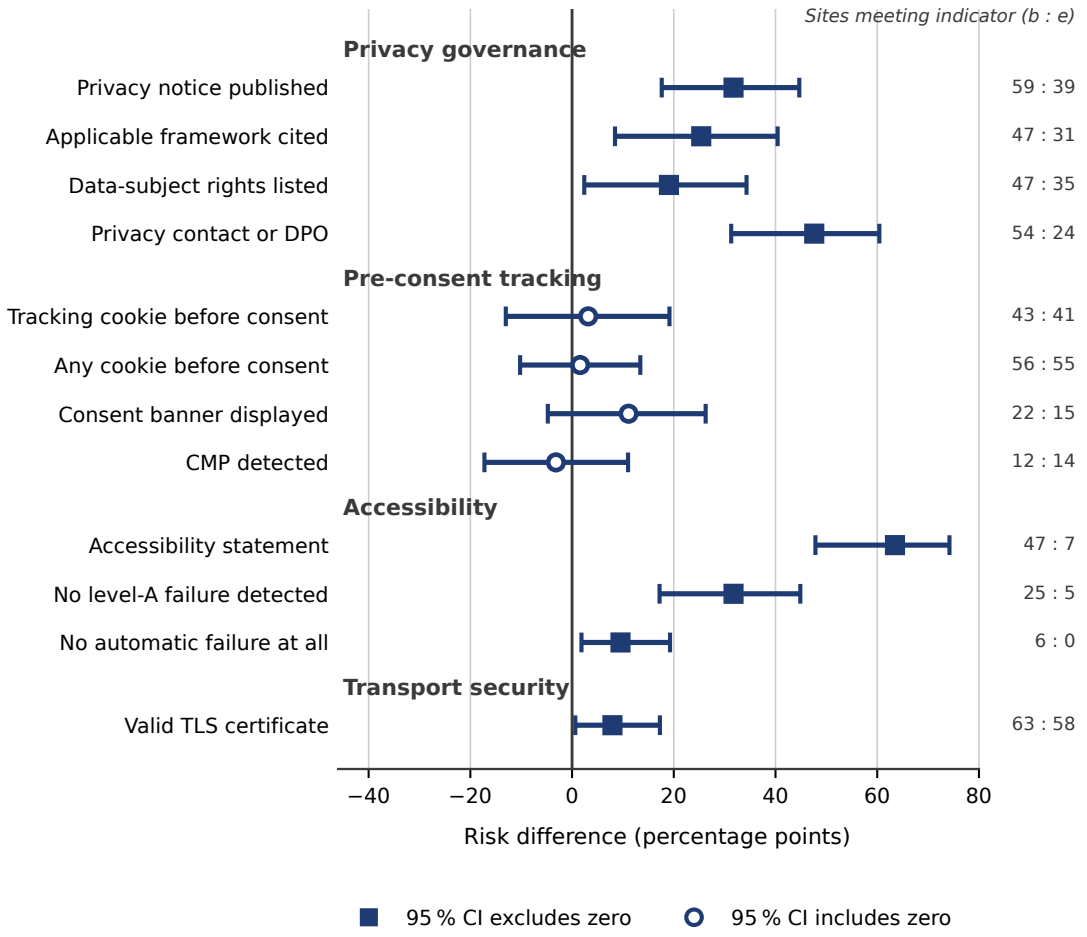


Figure 1. Risk differences between the benchmark group and the Ecuadorian census, with 95% Newcombe intervals, grouped by domain. Filled squares mark intervals that exclude zero; open circles mark intervals that include it. Intervals are not adjusted for multiple comparisons; the Holm-adjusted p values are given in Table 1. Counts of compliant sites out of 63 are given at the right. Positive values indicate a higher proportion in the benchmark group.

The regional breakdown of the benchmark group is given in Table 2. It should be read as descriptive: with subgroups of three to five institutions the Wilson interval around a proportion spans more than 40 percentage points, so no regional ranking can be supported by these data. Two patterns are nonetheless worth recording because they are large relative to that uncertainty. The Chinese institutions in the benchmark group are the least likely to publish a first-level notice, two of six, and none published an accessibility statement; the same absence of an accessibility statement holds for the five institutions grouped as Rest of Asia, so it is not a feature of the Chinese subgroup alone. Continental Europe cites a legal framework at nine of ten institutions, but the ten institutions are not homogeneous in the applicable instrument: two are Swiss and are subject to the revised Federal Act on Data Protection (Swiss Federal Assembly, 2020) rather than to the GDPR. Similarly, the three Canadian institutions are public bodies governed by provincial public-sector statutes (Legislative Assembly of British Columbia, 1996; Legislative Assembly of Ontario, 1990; National Assembly of Quebec, 1982), not by the federal Personal Information Protection and Electronic Documents Act (PIPEDA) (Parliament of Canada, 2000), which applies to commercial activity, so a single Canadian regime cannot be assumed.

The partial code affects these comparisons unevenly, and its effect is reported here so that it need not be inferred. Counting a partial cell as meeting the indicator instead of failing it leaves the privacy-contact difference unchanged at 47.6 percentage points, because that indicator admits no partial code, and moves the accessibility-statement difference by less than two points, from 63.5 to 61.9. It matters for the remaining three. The notice difference falls from 31.7 to 27.0 points and the rights difference rises from 19.0 to 27.0, in both cases without altering the conclusion drawn from them. The framework difference is the sensitive one: it rises from 25.4 to 41.3 points, because 11 of the 63 benchmark institutions name an instrument that does not govern them, against 1 of the 63 Ecuadorian institutions. Read either way the gap runs in the same direction, but the magnitude of the framework gap is conditional on how a misapplied instrument is treated, and no claim in this article rests on its precise size.

One further documentary indicator is recorded in Appendix C and is reported here rather than compared. A dedicated cookie policy, which is a published document and therefore distinct from the live banner measurement of Section 4.2, was found at 14 of the 63 benchmark institutions (22.2%). It is concentrated in the United Kingdom, at 5 of 7 institutions, and in continental Europe, at 4 of 10, and absent from all six Chinese institutions and from all five in the Rest of Asia group. It is excluded from Table 1 because the Ecuadorian matrix records no counterpart, so the indicator has no comparison group; it is retained in the appendix because it is the documentary complement of the live measurement and readers may wish to relate the two.

4.2. Pre-consent behaviour (RQ2)

Figure 2 reports the live measurements. Observed from Ecuador, the two groups behaved similarly. At least one cookie was set before any consent interaction at 88.9% of benchmark sites and 87.3% of Ecuadorian sites; tracking cookies specifically were set at 68.3% and 65.1%. The difference on the tracking indicator was 3.2 percentage points in favour of Ecuador, with a 95% interval of $[-13.0, +19.1]$ percentage points and an adjusted p of 1.000. A visible consent banner was, in both groups, a minority phenomenon (34.9% and 23.8%), and a consent-management platform was detected slightly more often in Ecuador (22.2%) than in the benchmark group (19.0%), a dif-

Table 2. Benchmark group by region: institutions meeting each documentary indicator out of the regional total. Percentages and intervals are omitted deliberately: with n between 3 and 24 they would suggest a precision the data do not have.

Region	n	Notice	Framework	Contact/DPO	Accessibility
United States	24	24	17	23	22
United Kingdom	7	7	4	5	7
Continental Europe	10	10	9	10	8
China	6	2	2	3	0
Hong Kong SAR	3	3	3	2	3
Rest of Asia	5	5	4	4	0
Australia	5	5	5	4	4
Canada	3	3	3	3	3
Total	**63**	**59**	**47**	**54**	**47**

ference well within sampling variation.

All figures in this section are computed under the extended cookie taxonomy of Appendix A, as is the whole of this article. Under the original list the Ecuadorian tracking count would be 40 of 63 rather than 41, a difference of one site, and the benchmark count would be unchanged; no conclusion in this section depends on that choice.

The absence of a detectable difference must not be read as a demonstration of equivalence. With 63 sites per group and a pooled observed proportion of $84/126 = 0.667$, the smallest difference this design could detect with 80% power at $\alpha = 0.05$ two-sided is 23.3 percentage points; detecting a true difference of the size observed here would require 3,461 sites per group. Two one-sided tests confirm the point directly: equivalence is rejected at a margin of ± 10 percentage points ($p = 0.208$) and at ± 15 ($p = 0.079$), and is only supported at a margin of ± 20 ($p = 0.023$), which is too wide to be substantively interesting. Demonstrating equivalence within ± 10 points would require, by two one-sided tests at $\alpha = 0.05$ with 80% power and the same pooled proportion, 275 sites per group if the true difference were exactly zero and 590 if it were of the size observed here, 3.2 points. The requirement depends critically on an assumption these data cannot fix, which is why both values are given. All four quantities in this paragraph are reproduced by `power_analysis.py` in the deposited code. The correct statement of this result is therefore that *the study found no evidence of a difference in pre-consent tracking, and had no power to establish that the groups are equivalent.*

A further and more important qualification concerns what a null result here can mean at all. Because all observations were taken from Ecuador, the benchmark figure conflates two possibilities that carry opposite legal readings: that European institutions do not condition their consent interfaces on the visitor's location and simply track everyone, or that they do condition them and therefore show a European visitor a different interface from the one recorded here. The present design cannot distinguish the two, and Section 5 sets out why the distinction matters and how it should be resolved.

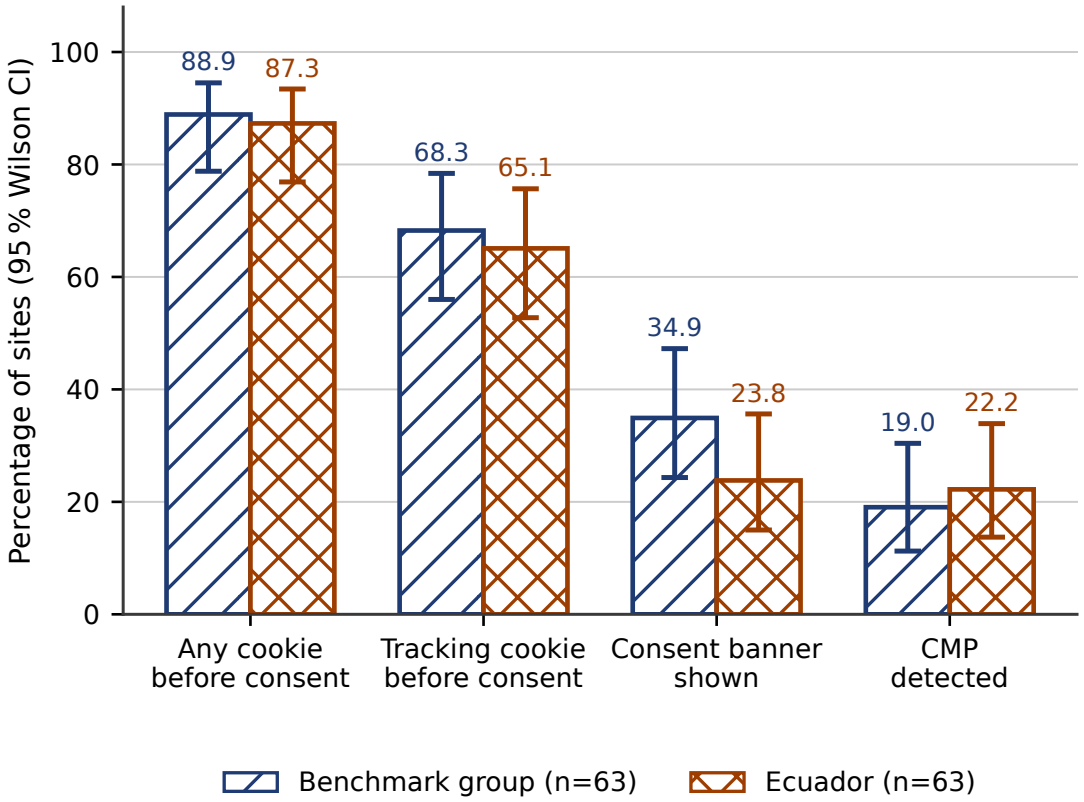


Figure 2. Cookie state recorded before any interaction with a consent interface, with 95% Wilson intervals. All measurements were taken from a single vantage point in Ecuador in August 2026; a European visitor would not necessarily observe the same behaviour on the European sites of the benchmark group.

4.3. Declared accessibility and transport security (RQ3, part one)

An accessibility statement or a set of accessibility tools was published by 47 of the 63 benchmark institutions (74.6%) and by 7 of the 63 Ecuadorian institutions (11.1%), the largest single difference in the study (odds ratio 23.50, adjusted $p < 0.001$). Concrete statutory duties plausibly account for part of this gap: 22 of the 24 United States institutions publish a statement, in a jurisdiction with the ADA and Section 508 (United States Congress, 1986, 1990), and 8 of the 10 continental European institutions do so under the European Accessibility Act (European Parliament and Council of the European Union, 2019). Ecuador's low figure is consistent with the regional lag documented over the past decade (Acosta-Vargas, González, & Luján-Mora, 2020; Campoverde-Molina et al., 2023).

Transport security was near-universal in both groups: a valid TLS certificate was present at all 63 benchmark sites and at 58 of the 63 Ecuadorian sites (92.1%; adjusted $p = 0.288$). We stress that this indicator measures confidentiality in transit only. A site with an impeccable certificate chain may transmit visitor data to any number of third parties, and the four pre-consent indicators of Table 1 show precisely that pattern in both groups.

The transparency section required of Ecuadorian public bodies by the Organic Law on Transparency and Access to Public Information (Ley Orgánica de Transparencia y Acceso a la Información Pública, LOTAIP) was present at 47 of the 63 Ecuadorian institutions (74.6%). It is reported here for completeness because it is recorded in Appendix D; it is not used in any comparison, since it has no counterpart in the benchmark group.

4.4. Measured WCAG conformance (RQ3, part two)

Table 3 reports the audit results and Figure 3 the distribution of failing nodes by conformance level.

Neither group performed well. No Ecuadorian site and only 9.5% of benchmark sites produced no automatic failure at any enabled level, and the median number of distinct rules violated was four in Ecuador compared with two in the benchmark group. The distinction that matters, however, is where the failures fall. In the benchmark group 19% of failing nodes were at level A and 60% at level AAA, the latter dominated by enhanced contrast; in Ecuador 45% were at level A. Excluding the AAA rules entirely and re-normalising over levels A and AA preserves the ordering, 46.0% compared with 61.1% at level A, which indicates that the qualitative finding is not an artefact of having enabled a single demanding rule.

Two readings that the aggregate figures invite are not supported by these data, and both are worth ruling out explicitly because a benchmarking design encourages them. It is *not* the case that no institution is free of automatic failures: six benchmark institutions were. And it is *not* the case that the benchmark group fails only at the level of excellence: 60.3% of benchmark sites produced at least one level-A failure, so the majority of the reference group does not clear the basic tier either. What the data support is narrower and still worth reporting: the two groups differ in the *distribution* of their failures, not in the presence of failures.

The single clearest instance is success criterion 2.4.4, which requires that every link have text a screen reader can announce; 81.0% of Ecuadorian sites and 23.8% of benchmark sites violated it. Insufficient contrast (1.4.3) followed the same pattern, 79.4% compared with 25.4%. The most affected principle in both groups is perceivability,

Table 3. Automatically detectable WCAG results (axe-core 4.13, August 2026, home page only). Percentages are of sites unless stated otherwise. The rule set is specified in Section 3.5.

Measure	Benchmark	Ecuador
Median distinct rules violated per site	2	4
Sites with no automatic failure at any level	9.5%	0.0%
Sites with no automatic level-A failure	39.7%	7.9%
Links without an accessible name (2.4.4, level A)	23.8%	81.0%
Insufficient contrast (1.4.3, level AA)	25.4%	79.4%
Images without a text alternative (rule `image-alt`, success criterion 1.1.1, level A)	15.9%	30.2%
Distribution of failing nodes by level, all rules enabled		
Level A	19%	45%
Level AA	22%	29%
Level AAA	60%	26%
Same distribution with AAA rules excluded and re-normalised		
Level A	46.0%	61.1%
Level AA	54.0%	38.9%

Note: the median is of distinct rules violated, not of failing nodes. Level percentages are of each group's total failing nodes and are rounded to the nearest integer, which is why the benchmark distribution sums to 101%. The re-normalised block is computed from the unrounded node counts and not from the rounded percentages above it, which is why 19 and 22 give 46.0 and 54.0 rather than 46.3 and 53.7.

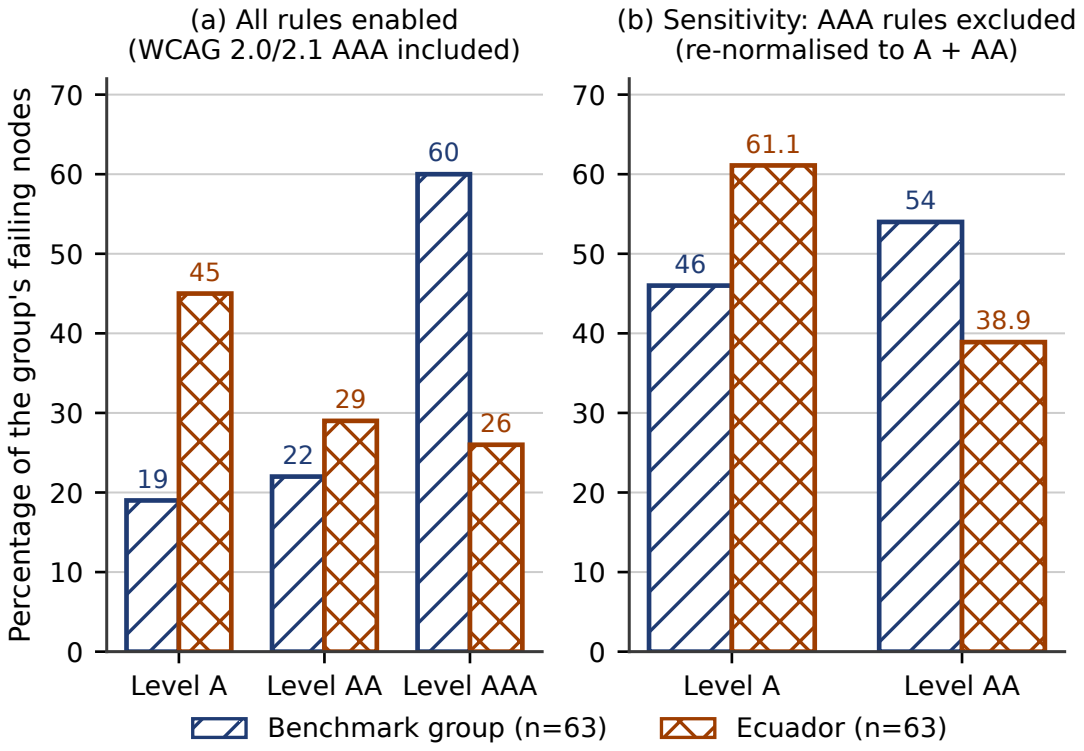


Figure 3. Distribution of failing nodes by WCAG conformance level. Panel (a) uses the full rule set of this study, which includes the level-AAA rules of WCAG 2.0 and 2.1. Panel (b) excludes those rules and re-normalises over levels A and AA, showing that the ordering of the two groups at level A is preserved when the enhanced-contrast rule is not counted.

through contrast; Ecuador adds an operability deficit through links and controls that is marginal in the benchmark group. These are the same dominant error types reported for other public-sector portals (Aljedaani, 2025; Iniesto & Rodrigo, 2025).

Two cautions apply to the level-based measures. First, the distribution in Figure 3(a) is computed over failing *nodes*, which are not independent within a site: a single template with several thousand low-contrast elements can dominate its group's distribution. Second, the distribution is compositional, so a group with more level-A failures must, arithmetically, show a smaller AAA share. Both are reasons to treat the level distribution as descriptive; the site-level measures in the upper half of Table 3 are the inferentially safe ones, and they point in the same direction.

4.5. What exhaustive manual coding finds, and the automated audit does not (RQ3, part three)

Across the fifteen sites of the subsample, no site satisfied all three success criteria and ten satisfied none of them. Success criterion 2.4.4 was not satisfied at any of the fifteen, 1.1.1 at twelve, and 1.4.3 at twelve. Every row that the ACT rules refer to human judgement and that could still decide a verdict has been resolved, so no cell of the fifteen-by-three matrix is left open and the counts below are complete rather than lower bounds. Table 4 gives the site-by-site detail alongside the automated result for the same page.

The comparison with the automated audit is the substance of this subsection. Of the 45 site-by-criterion cells, exhaustive coding under the ACT rules finds failures in 39. In 23 of those 39 cells, that is in 59 % of them, axe-core had reported no violation of the corresponding success criterion at that site. The shortfall is largest, not smallest, in the criterion whose machine test is most familiar: for 1.1.1 the automated audit reported nothing at 9 of the 12 failing sites, flagging that criterion only at UTI, UTPL and NUS; for 2.4.4 at 8 of 15; and for 1.4.3, which an automated checker computes arithmetically, at 6 of 12.

The first coding, whose SC 1.1.1 judgements were resolved with assistance, would also fail ECOTEC and UC Berkeley on that criterion; on that reading the count would be 25 of 41 cells, 61 % rather than 59 %. Both differences turn on the same question, whether an accessible name that describes something other than the image it labels serves a purpose equivalent to it, and both are recorded element by element with the coders' written justifications in the deposit. The finding of this subsection is the order of magnitude of the shortfall and does not depend on which of the two readings is taken.

Six of the fifteen sites showed no level-A failure at all under the automated audit. All six fail 2.4.4, a level-A criterion, and four of them also fail 1.1.1, so in every case examined the appearance of meeting the minimum did not survive an element-by-element reading. Four of the six, Manchester, Northwestern, UC Berkeley and UCL, produced no automated violation at any conformance level whatsoever, and these four are among the six benchmark institutions, 9.5 % of that group, that Section 4.4 reports as free of automated failures. Manchester fails all three criteria examined, Northwestern and UCL fail 1.1.1 and 2.4.4, and UC Berkeley fails 2.4.4.

None of this impeaches the automated instrument, which is not designed to certify conformance. It quantifies, on the pages this study audits, what Fischer et al. (2025) establish in general: automated checkers address a minority of the WCAG success criteria, so passing an automated check is not evidence that a site is accessible. The

contribution here is the size of that gap where it bears on the present argument. For the three criteria most central to it, nearly six of every ten site-criterion failures are invisible to the automated instrument, and the category "no level-A failure detected", which covers 30 of the 126 sites in this census, did not survive examination in a single one of the six cases where it was examined.

Table 4. Automated result and exhaustive manual coding on the fifteen sites of the validation subsample. The three numeric columns are counts of failing nodes reported by axe-core 4.13.0 by conformance level on 14 August 2026. "Flagged" lists which of the three success criteria the automated audit reported at all. The three right-hand columns give the verdict of the second, wholly human manual coding of 9–11 September 2026 (Section 3.6), in which a criterion is not satisfied at a site if any applicable element fails; no cell is left open. Bold marks the 23 cells in which the manual coding finds failures and the automated audit had reported no violation of that criterion. Success criteria 1.1.1 and 2.4.4 are level A; 1.4.3 is level AA.

Site	Group	A	AA	AAA	Flagged	1.1.1	1.4.3	2.4.4
ECOTEC	Ecuador	0	6	27	1.4.3	yes	no	**no**
IAEN	Ecuador	2	1	2	1.4.3	**no**	no	**no**
ULVR	Ecuador	11	2	0	1.4.3, 2.4.4	**no**	no	no
UNESUM	Ecuador	8	0	0	2.4.4	**no**	**no**	no
UPS	Ecuador	10	0	0	2.4.4	**no**	**no**	no
USECIPOL	Ecuador	0	0	4	none	**no**	**no**	**no**
UTI	Ecuador	9	10	1	1.1.1, 1.4.3, 2.4.4	no	no	no
UTPL	Ecuador	9	1	25	1.1.1, 2.4.4	no	**no**	no
Cornell	Benchmark	1	0	5	none	yes	**no**	**no**
HKUST	Benchmark	14	6	0	1.4.3, 2.4.4	**no**	no	no
Manchester	Benchmark	0	0	0	none	**no**	**no**	**no**
NUS	Benchmark	24	21	15	1.1.1, 1.4.3, 2.4.4	no	no	no
Northwestern	Benchmark	0	0	0	none	**no**	yes	**no**
UC Berkeley	Benchmark	0	0	0	none	yes	yes	**no**
UCL	Benchmark	0	0	0	none	**no**	yes	**no**
Sites not satisfying the criterion						12 / 15	12 / 15	15 / 15
Of those, criterion not flagged by the automated audit						9	6	8

4.6. Do the two commitments co-occur? (RQ4)

Across all 126 sites, the Spearman correlation between the number of failing accessibility nodes and the number of pre-consent tracking cookies is 0.05. That coefficient has a 95% confidence interval of $[-0.13, +0.22]$ and $p = 0.578$, and the design could only have detected a correlation of about 0.25 with 80% power. The pooled coefficient is therefore compatible with no association and with a weak positive association alike, and it cannot support a claim of independence.

The pooled coefficient is also the wrong estimand, because it mixes two populations whose marginal distributions differ on both variables. Table 5 gives the within-group cross-classification of the two dichotomised measures. Within the benchmark group the association is, if anything, slightly negative (odds ratio 0.54, Fisher $p = 0.408$); within the Ecuadorian group it is essentially absent (odds ratio 1.27, Fisher $p = 1.000$); the Mantel–Haenszel estimate adjusted for group is 0.67. None of these is distinguishable from no association at this sample size.

The defensible conclusion from RQ4 is a negative one, and it is worth stating precisely because the temptation is to overstate it. These data provide no evidence that institutions which invest in accessibility also restrain tracking, and no evidence of the reverse. They are not sufficient to establish that the two are independent; establishing

Table 5. Cross-classification of automatically detectable level-A conformance against pre-consent tracking, within each group. OR: odds ratio of conformance given restraint in tracking, with 95% interval.

Group	A^+T^-	A^+T^+	A^-T^-	A^-T^+	OR [95% CI]	p
Benchmark	6	19	14	24	0.54 [0.17, 1.68]	0.408
Ecuador	2	3	20	38	1.27 [0.20, 8.21]	1.000

Note: A^+ no automatically detectable level-A failure, A^- at least one; T^- no tracking cookie before consent, T^+ at least one. p from the two-sided Fisher exact test. In Ecuador, 2 institutions of 63 met both criteria. Under independence, $63 \times 0.079 \times 0.349 \approx 1.7$ would be expected, so this figure is exactly what chance predicts and carries no information about coupling.

that would require either a substantially larger sample or a continuous, better-powered pair of measures.

4.7. Whose location determines protection? (RQ5)

The four-vantage replication yielded 121 of the 126 sites measured successfully in all three passes at all four vantage points. The five sites that fall out are named here, because a paired design cannot impute them and the reader is entitled to know which they are. Each failed to load on at least one pass, in every case with a connection timeout or with the 45 s navigation limit exceeded, that is, because the server did not respond rather than because the audit was interrupted. They are Université Paris-Saclay, which failed on one British pass; ESPOL, on two German passes; UCSG, on two Ecuadorian passes; UDET, on one German and all three British passes; and UNEMI, on all three Ecuadorian passes and one United States pass. Four of the five are Ecuadorian, which is itself a finding about the availability of those portals and is taken up in Section 5.6. The 121 sites retained comprise 62 of the benchmark group and 59 of the Ecuadorian census. The three control quantities behaved as required: the mean number of WCAG violations per site was 3.53, 3.51, 3.54 and 3.50 at the Ecuadorian, German, British and United States vantage points respectively, a spread of 1.1% between the extremes, confirming that the pages themselves rendered identically. Between-pass agreement on the tracking indicator, computed pairwise within each vantage point over the 121 sites, ranged from 95.9% to 100% across the twelve pairs. That range describes the stability of the measure over the passes actually run; it does not establish that three passes are sufficient in general, and the majority rule is used precisely because single passes disagree at a small number of sites.

Pre-consent tracking, by contrast, differed by visitor location. It was observed at 82 of the 121 sites from Ecuador (67.8%, 95% CI 59.0–75.4), at 79 from the United States (65.3%), at 76 from the United Kingdom (62.8%) and at 73 from Germany (60.3%, 51.4–68.6). Cochran's Q over the four vantage points gives $Q = 15.88$, 3 df, $p = 0.0012$.

The paired comparisons in Table 6 locate that difference, and the multiplicity policy of Section 3.9 is applied to the six of them as a family. Ecuador differs from Germany, with 9 discordant pairs all in the same direction (exact $p = 0.0039$; Holm-adjusted $p = 0.023$), and this contrast is the finding of the section. It is not the contrast that Section 3.7 identified as free of any residential-versus-data-centre artefact, which is Germany against the United States: at site level that contrast points the same way, with 7 of its 8 discordant pairs favouring the United States, but does not survive correction (raw $p = 0.070$, adjusted $p = 0.281$), whereas at vendor level it reproduces the

suppression in full, as shown below. The Ecuadorian endpoint is examined separately in Section 5.6. The difference between Ecuador and the United Kingdom does not survive correction within the family (raw $p = 0.031$, adjusted $p = 0.156$) and is reported as suggestive only. No other pair is distinguishable: the smallest adjusted value among the remaining four is 0.281, for the United States against Germany.

The conclusion does not depend on that inclusion rule. Admitting sites with at least two valid passes per vantage point ($n = 122$) leaves every count and test of Table 6 unchanged, and admitting sites with a single valid pass ($n = 124$) strengthens the Ecuador–Germany contrast to ten discordant pairs, all in the same direction (Holm-adjusted $p = 0.012$). The strictest rule is reported because it is the most conservative.

Three tracking figures for Ecuador appear in this article and are not interchangeable, so the relation between them is stated here rather than left to the reader. Table 1 reports 41 of the 63 Ecuadorian sites and 43 of the 63 benchmark sites, that is 84 of the 126 in all (66.7%), from a single pass on 14 August 2026. The present section reports 82 of the 121 sites observed successfully at every vantage point (67.8%), consolidated over three passes on 15 August 2026. Both use the extended taxonomy, and the two figures differ in site set, consolidation rule and date, and neither supersedes the other: the first supports the between-group comparison, the second the within-site comparison across vantage points.

Table 6. Exact McNemar tests for pre-consent tracking between vantage points ($n = 121$ sites observed at all four). *a*: sites tracking at both vantage points; *b*: sites tracking at the first but not the second; *c*: the converse; *d*: sites tracking at neither. Only discordant pairs carry information. p_{H}: Holm-adjusted over the six comparisons of this family.

Comparison	*a*	*b*	*c*	*d*	**Discordant**	*p*	$p_{\mathbf{H}}$
Ecuador vs Germany	73	9	0	39	9	0.0039	0.023
Ecuador vs United Kingdom	76	6	0	39	6	0.031	0.156
United States vs Germany	72	7	1	41	8	0.070	0.281
United Kingdom vs Germany	73	3	0	45	3	0.250	0.750
Ecuador vs United States	79	3	0	39	3	0.250	0.750
United States vs United Kingdom	75	4	1	41	5	0.375	0.750

Note: rows are ordered by raw *p*. The marginal totals implied by the table are 82 sites tracking from Ecuador, 79 from the United States, 76 from the United Kingdom and 73 from Germany, which are the counts reported in the text.

The mechanism becomes visible once the cookies are attributed to the vendor that sets them. Figure 4 shows the share of sites at which each vendor set at least one cookie before any consent interaction. On 15 August 2026 three vendors set no cookies at all at the German and British vantage points while setting them at the Ecuadorian and United States ones, on the same pages under the majority rule over three passes: Microsoft, whose Clarity, advertising and Universal Event Tracking (UET) cookies appeared at 14 sites from Ecuador and 14 from the United States and at none from Germany or the United Kingdom; TapAd, at 4 and 4 sites and at none; and StackAdapt, at 2 and 2 sites and at none, a count too small to support anything beyond its own description. The LinkedIn Insight Tag followed the same pattern in attenuated form, at 14 and 14 sites compared with 3 and 7. The other large vendors behaved differently from one another: Google Analytics fell moderately, from 71 and 69 sites from Ecuador and the United States to 65 and 63 from the United Kingdom and Germany; Meta fell from 24 to 19 and 15; and Adobe/Tealium set cookies at 4 sites from every vantage point. These counts describe conduct on one date; the layer at which each

difference is decided is examined below with the network records of the following day.

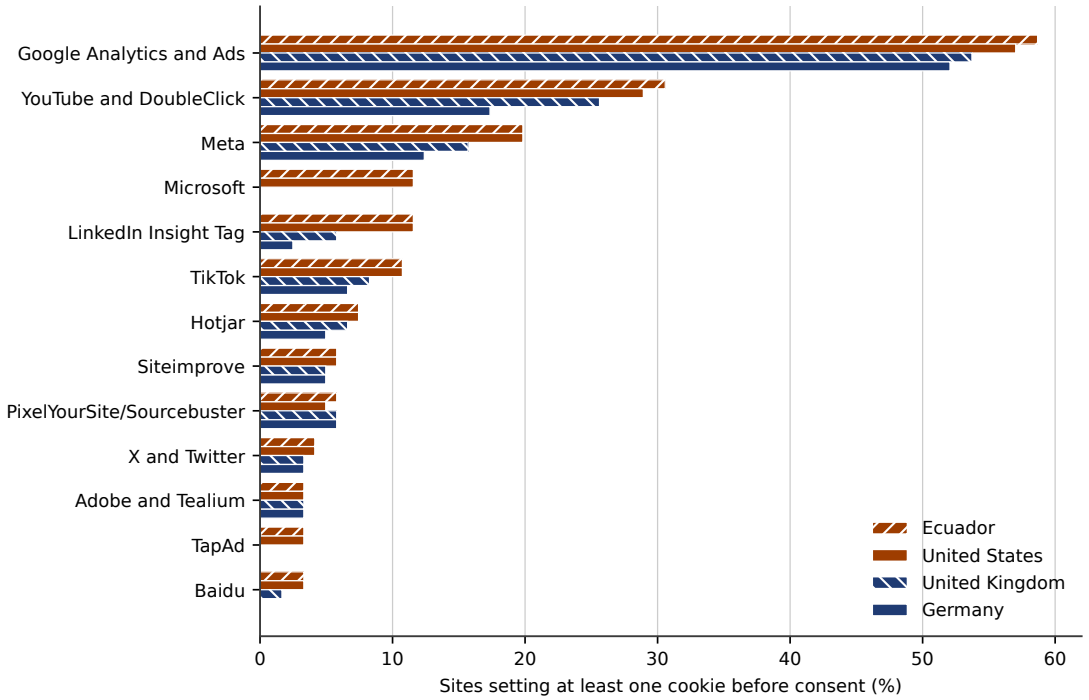


Figure 4. Share of sites at which each advertising vendor set at least one cookie before any consent interaction, by vantage point ($n = 121$ sites, majority of three passes, August 2026). Colour marks the perimeter observed for vendor cookies, not a legal boundary; Table 7 shows at which layer it is applied: navy for the two vantage points at which vendor cookies were suppressed, Germany and the United Kingdom, and ochre for the two at which they were not, Ecuador and the United States. That this commercial perimeter includes the United Kingdom, which is outside the European Economic Area, is a finding of Section 4.7 and not a premise of the figure. Vendors are ordered by their prevalence as seen from Ecuador. Thirteen of the twenty-one families of the taxonomy of Appendix A are shown. Six are omitted because they were present at fewer than four sites from Ecuador, namely Privacy Sandbox, StackAdapt, Snapchat and ContentSquare at two sites each and Pinterest and Matomo at one; the remaining two, Quantcast and LiveIntent, set no matching cookie at any site from any vantage point and are therefore absent from the figure as a result of the measurement rather than of a display choice.

Attribution by cookie name says whose cookie is missing, not who withheld it. The network records of 16 August, described in Section 3.8, answer that question, and Table 7 gives the result for the vendors whose cookies fell most. Microsoft again set cookies at 14 sites from Ecuador and from the United States and at none from Germany, the United Kingdom or Switzerland. At 13 of those 14 sites the browser in Frankfurt still sent requests to Microsoft's domains, no response carried a `Set-Cookie` header and no Microsoft cookie appeared, so the cookie was withheld by Microsoft's own code. At the fourteenth, the University of Manchester, the request was never sent from Frankfurt or London and the site runs a consent-management platform, so there the site decided. Eleven of the 13 sites run no consent-management platform and no consent default for Microsoft's or Google's tags was detected in their code, so nothing on those pages told Microsoft's code what to do; at the other two, Carnegie Mellon and UIDE, a consent-management platform is present, and a signal passed through it to Microsoft cannot be ruled out. The 13 sites are Ecuadorian, United States, Canadian, Australian and Hong Kong universities, none of them established in the European Union, and on the evidence of the requests none of them decided what a visitor in Frankfurt would receive. TapAd behaves differently. Its cookies were set at four sites

from Ecuador and from the United States, but from the three European vantage points no request to TapAd's domain was sent at all, so the decision was taken upstream of TapAd, by the site or by another script that would otherwise have called it; these data do not say which. Meta and LinkedIn show both patterns at once: of the 25 sites with Meta cookies from Ecuador, 16 kept them in Frankfurt, 8 lost them although the request was sent and 1 lost them because it was not. The perimeter is therefore applied mostly, but not only, by the vendors, and its edge differs from one vendor to the next. The United Kingdom, which is outside the European Economic Area, lies inside it for most vendors, while TikTok and Baidu set more cookies for British visitors than for German ones on 15 August, at 10 sites against 8 and at 2 against 0.

Table 7. Layer at which each vendor's cookie is withheld, 16 August 2026 ($n = 121$ sites, majority of three passes). For the sites at which the vendor set a cookie for a visitor in Ecuador, each cell gives, at the vantage point shown, the sites at which the cookie was still set, the sites at which the request to the vendor was sent but no cookie resulted (decided by the vendor), and the sites at which the request was not sent (decided upstream of the vendor).

Vendor	Sites, Ecuador	Germany	United Kingdom	Switzerland
Microsoft	14	0 / 13 / 1	0 / 12 / 2	0 / 13 / 1
LinkedIn	13	4 / 7 / 2	3 / 7 / 3	2 / 9 / 2
Meta	25	16 / 8 / 1	15 / 8 / 2	11 / 13 / 1
TapAd	4	0 / 0 / 4	0 / 0 / 4	0 / 0 / 4
StackAdapt	2	0 / 1 / 1	0 / 0 / 2	0 / 1 / 1
Google Analytics	69	61 / 3 / 5	61 / 3 / 5	55 / 4 / 10

Note: cells read set / withheld by the vendor / withheld upstream. From the United States every vendor set its cookie at the same sites as from Ecuador, except Meta at one site and Google Analytics at two, where the request was not sent. Requests are attributed to a vendor by domain, as listed in the deposited instrument.

The site-level result of 15 August reappears on 16 August in the same direction. Of the 121 sites, tracking cookies were set at 81 from Ecuador, 80 from the United States, 75 from Germany and from the United Kingdom, and 70 from Switzerland (Cochran's $Q = 23.18$, 4 df, $p = 0.00012$). All six sites that differed between Ecuador and Germany tracked from Ecuador and not from Germany (exact $p = 0.031$), but in this family of ten comparisons that contrast does not survive Holm adjustment ($p_{\mathrm{H}} = 0.25$); the contrasts that do are Ecuador and the United States against Switzerland, with 11 and 10 discordant pairs all in one direction ($p_{\mathrm{H}} = 0.0098$ and 0.018). The five sites without valid passes at every vantage point on 16 August, UCSD, UDET, UG, UPACIFICO and UTN, are not the same five as on 15 August.

One caveat applies. Cookies not attributable to any advertising vendor also fell at the European and British vantage points, by 26.1% and 21.8% relative to Ecuador, whereas the United States vantage point differed by only 0.8%. That residual is smaller than the 43.6% and 37.4% fall in vendor cookies and it does not affect the rendering controls, but part of it is likely to be the Cloudflare bot-management cookie and the cookies set by consent platforms themselves, whose behaviour also varies with the visitor's region. The residual is reported here rather than adjusted away.

5. Discussion

5.1. Interpretation

Three findings survive the analysis of Section 4, and they are of different kinds.

The first is a governance gap that is large, robust to multiple-comparison correction and consistent across four indicators. The Ecuadorian shortfall is not principally the absence of documents but the absence of the elements that make a document actionable: a named instrument, an enumeration of rights and, above all, an identified route through which a person can exercise them. The 47.6-point gap on the privacy contact is the single most consequential number in the study, because a right without a channel is not exercisable. That benchmark institutions perform well here is not evidence of superior ethics; it is the expected consequence of legal regimes that impose the duty explicitly, and it suggests that the same is achievable in Ecuador as the LOPDP and its regulation move into their enforcement phase (Asamblea Nacional del Ecuador, 2021; Presidencia de la República del Ecuador, 2023).

The second is an accessibility gap whose interesting property is its shape. Both groups fail; the majority of the benchmark group fails at the basic tier too. What differs is that Ecuadorian failures concentrate at level A, where the remedies are inexpensive and well understood, principally accessible link text, which alone accounts for 627 of the 990 level-A failing nodes at Ecuadorian sites, and where the consequences for users of assistive technology are most severe. This is the most immediately actionable finding of the study and the one that requires no new legislation to act on.

The third is a null result on pre-consent tracking that must be reported as such. Observed from Ecuador, roughly two thirds of the institutions in both groups set tracking cookies before any consent interaction. The study found no difference; it also had no power to establish equivalence, and Section 4.2 states the bound explicitly. The finding that survives is therefore not "there is no gap" but "the gap, if any, is not large, and it is certainly not in the direction that the reputational ordering of the two groups would predict".

A fourth observation follows from the absence of association between the two barriers, and it bears on the framework with which this article opened. Section 2.1 motivated measuring accessibility and pre-consent tracking together on the ground that both are costs a service imposes on the people obliged to use it, and that universal design treats such costs as properties of the service rather than of the user. If that shared root were also an organisational one, institutions that invest in one form of access would tend to restrain the other. The data give no evidence that they do. The conceptual unity of the two barriers is therefore not matched by an institutional unity, at least not at the level of the home page and not in these two populations. The reading we favour is not that the framework fails but that it is prescriptive rather than descriptive: it identifies what an inclusive service would have to do, and the absence of correlation measures the distance between that requirement and current practice, in which accessibility and tracking are governed by different offices, different suppliers and, as Section 5.2 shows for the second of them, sometimes by no one within the institution at all. This is a hypothesis these data cannot test, and it is offered as such; what the data do establish is that a single institutional commitment cannot be inferred from either barrier alone, which is a practical consequence for anyone who would audit only one of the two.

5.2. Territorial scope, and who actually enforces it

One explanation of the benchmark group's behaviour suggests itself immediately, and it is wrong twice over: that the protection of the GDPR stops at the Union's border, so that a student consulting a European university's portal from Ecuador is as exposed as on a domestic site. It is wrong in law, and the four-vantage replication shows that

it is also wrong in fact, though not in the direction that framing would predict.

The legal point first. Article 3(1) of the GDPR applies to processing in the context of the activities of an establishment of a controller in the Union *regardless of whether the processing takes place in the Union or not*, and the European Data Protection Board (EDPB) has confirmed that the location of the data subject is irrelevant to that limb (European Data Protection Board, 2019; European Parliament and Council of the European Union, 2016). A German or French university that sets marketing cookies in the browser of a visitor located in Ecuador is therefore acting within the material and territorial scope of the Regulation. Article 3(2) extends the Regulation to controllers not established in the Union that offer goods or services to data subjects *who are in* the Union, so that limb keys on the data subject's location rather than the controller's. The narrower and defensible point concerns a different instrument: Article 5(3) of the ePrivacy Directive, as transposed nationally, is articulated around the terminal equipment of a user, and the EDPB's guidance on its technical scope is the appropriate reference (European Data Protection Board, 2024; European Parliament and Council of the European Union, 2002).

The measurement then shows something the institutional framing does not anticipate. Protection does vary with the visitor's location, and substantially, but for the largest vendor involved it is not the universities that vary their behaviour. At 13 of the 14 sites where Microsoft set cookies for a visitor in Ecuador, a visitor in Frankfurt still sent requests to Microsoft's servers and received no cookie, on Ecuadorian, United States, Canadian, Australian and Hong Kong university sites alike. The institution is a bystander to this: it embeds a tag, and the logic inside the tag decides what the visitor receives. For other vendors, TapAd among them, the request is withheld before it reaches the vendor, and the network records cannot say whether the site or another third party withheld it.

Three consequences follow, and together they are the most original contribution this study can make.

The first is that, where the vendor decides, compliance is being executed at the wrong layer, and by an actor whose legal position is stronger than the institutional framing suggests. A vendor that sets its own cookies, for its own advertising and measurement purposes, and that decides on the basis of its own country list whether to set them, is not acting on documented instructions from the university: it is determining purposes and means. In *Fashion ID* the Court of Justice held that the operator of a site embedding a third-party plug-in is a joint controller with the provider in respect of the collection and transmission of visitors' data, and that the operator's lack of access to those data is irrelevant to that classification (Court of Justice of the European Union, 2019); the EDPB takes the same line for embedded third parties acting for their own purposes (European Data Protection Board, 2021). The judgment was given under Directive 95/46/EC, which was the applicable law at the time, and the attribution it establishes has two tiers: joint controllership covers the collection and transmission, while the provider is sole controller of everything it does with the data thereafter. Read this way the accountability problem is not only that the university cannot demonstrate compliance for conduct it does not observe. It is that a second controller, bearing its own obligations under Articles 26 and 5(2), operates an unpublished country list that no supervisory authority is examining.

The second is that the boundary is commercial, not legal. The United Kingdom left the European Economic Area but most vendors still place it inside the protected perimeter, while TikTok and Baidu do not. A person's protection therefore depends on which country lists a set of private firms happen to maintain, and those lists are

neither published, nor harmonised, nor subject to the review that a legal instrument would receive.

The third is distributive, and it is the point that connects this finding to the concern of universal access with which this article began. The population that receives no protection is the one located outside the vendors' perimeter. In this measurement the perimeter left out both Ecuador and the United States: a visitor in Cuenca or Miami received Microsoft cookies on 13 university home pages that set none for a visitor in Frankfurt, London or Zurich. Whether it also leaves out other regions was not measured, and Section 6 sets this out as future work. When the page is that of an Ecuadorian institution, the applicant who receives less protection is also less likely to find a privacy contact on it, 38.1% against 85.7% in the benchmark group, so the user with more exposure has fewer ways to ask about it. The asymmetry runs in the direction that compounds existing inequality, and it is invisible to any monitoring exercise conducted from inside the perimeter, which is where regulators, auditors and most researchers sit. This is precisely the exclusion by geopolitical and economic position that Abascal et al. (2016) argue universal accessibility must address alongside exclusion by ability, and it appears here in an unusually tractable form: the perimeter is a list of countries, so the excluded population can be enumerated and the boundary can be measured.

5.3. What users can see and do

Both barriers act before a user can make any choice. Pre-consent tracking happens on arrival, before a banner, if there is one, can be answered, and for most of the vendor cookies examined here the decision to track was taken by a third party whose code the visitor cannot see and the institution does not configure. Research on privacy behaviour locates the user's means of response in privacy literacy and self-efficacy, which depend in part on how easily people reach privacy information (Epstein & Quinn, 2025; Meier & Krämer, 2025). The documentary gap of Section 4.1 bears on exactly that access: a notice that names no instrument, lists no rights or gives no contact leaves an applicant without the information from which literacy is built. Presentation matters as well, since the design of a consent request changes how attentive users are to what they disclose (Lindegren et al., 2021) and a visual presentation of a privacy policy raises trust and lowers perceived intrusion (Magrizos et al., 2026); a cookie set before any request is shown makes both irrelevant, because the disclosure has already happened. Users in different countries weigh privacy differently (Li et al., 2022), and the present results add that the same page also treats them differently. On the accessibility side, links without an accessible name are the largest single source of level-A failures at Ecuadorian sites, and they affect users of screen readers directly. That failures of this kind persist is consistent with what is known about the people who build these pages, who know the guidelines poorly and find them hard to engage with (Cao & Loiacono, 2022; Spyridonis & Daylamani-Zad, 2021), which places part of the remedy in developer education rather than in audit alone.

5.4. Implications for practice

For Ecuadorian institutions, the ordering implied by these results is unusually clear. The first action is to identify a privacy contact by role and address on the first-level notice; it is the cheapest intervention in the list and it closes the largest gap. The

second is to name the applicable instrument and enumerate the rights the LOPDP confers, together with the procedure to exercise them, replacing generic references to "applicable legislation". The third is to remediate the level-A accessibility failures, beginning with link text, which alone accounts for 63% of the level-A failing nodes observed at Ecuadorian sites; contrast failures, by contrast, are level AA and come after. The fourth, and the least urgent on the evidence collected here, is the consent interface: adding a banner without configuring it to block before consent changes the measured indicator without changing the user's exposure, a substitution that the measurement literature has documented repeatedly (Bollinger et al., 2022; Sanchez-Rola et al., 2019) and that the present data cannot rule out in either group.

For institutions in the benchmark group, the finding that 60.3% of them produce at least one level-A failure, and that two thirds set tracking cookies before consent when observed from outside the Union, is the relevant one. Reputational standing is not evidence of either accessibility or privacy performance. A further implication follows from Section 4.7 and applies to every institution in both groups: an administrator who tests their own site from their own office sees only what their own location entitles them to see. Verifying what an international applicant experiences requires testing from that applicant's region, which is neither difficult nor expensive and is, on this evidence, not being done.

5.5. Implications for policy

Two implications follow for regulators. First, for the Ecuadorian data protection authority, the indicators that separate the two groups most sharply are exactly those that are verifiable at low cost from outside the institution, so a periodic supervised self-assessment against a published matrix is administratively feasible; the codebook in Appendix A is offered for that purpose. The accompanying risk is well known and should be anticipated: an indicator that becomes a target is satisfied at the surface, and the cookie banner is the clearest instance of an indicator that can be met without any change in behaviour.

Second, the pattern documented in Section 4.7 concerns European supervisory authorities rather than Ecuadorian ones, and it raises a question that supervision as currently practised does not ask. Where a university's compliance with Article 5(3) is delivered de facto by the geolocation logic of a vendor that is itself a controller, neither party can demonstrate accountability under Article 5(2) for the outcome: the university neither configured nor observes it, and the vendor's country list is not published. Supervisory testing conducted from inside the perimeter will find that controller compliant. We suggest that audits of consent behaviour be performed from outside the European Economic Area as a matter of routine, and that the country lists operated by major tag vendors be treated as a supervisable object in their own right.

5.6. Threats to validity

Construct validity. Pre-consent tracking is measured uniformly across twelve jurisdictions whose obligations differ radically; in the United States there is no general prior-consent duty, and Ecuadorian law does not incorporate an equivalent of Article 5(3) of the ePrivacy Directive. Appendix B maps each indicator to the concrete obligation, if any, in each jurisdiction, and the results in Section 4.2 are stated as observed conduct, not as compliance findings. "No level-A failure detected" is a statement

about the machine-testable subset and not a conformance claim (Fischer et al., 2025). Transport security is reported separately from data protection for the reason given in Section 4.3. Cookie-based measurement also misses `localStorage`, `IndexedDB`, fingerprinting, cookieless pixels and server-side tagging, so the tracking figures reported here are lower bounds.

Internal validity. The benchmark group is an elite stratum, not a sample of the world, so resource and jurisdiction effects are fully confounded; no third stratum was measured. Confounders that could be recorded were not, in particular budget, enrolment, site language, content management system and hosting provider, and the last of these is a plausible source of non-independence within the Ecuadorian group, where shared templates are common. The institution type recorded in Appendix D, public, co-financed or self-financed, is not analysed here.

Reliability. The documentary matrix was compiled with the assistance of a generative artificial intelligence system and then coded and reviewed with the participation of the three authors and one external collaborator, whose individual codings were not retained, so no inter-coder reliability coefficient can be reported. The verification substudy of Section 3.4 found two errors in 24 re-examined cells; because those cells were selected purposively, the figure is neither an upper nor a lower bound on the error rate of the full matrix, and it is reported only as evidence that the rate is not negligible. The coding of items requiring judgement across several languages, among them Chinese, Japanese, Korean, German, French and Swedish, is the most exposed part of the design. The asymmetry with the live measurements is itself a limitation: geolocation was verified 84 times on 15 August and 105 times on 16 August, and the cookie state consolidated over three passes, whereas no documentary cell has a retained individual coding, and only 24 of them were re-examined by hand against the live site. Partial codes were treated as non-compliant, a convention that penalises institutions with segmented notices; Section 4.1 reports the effect of the alternative convention. Codes for non-verifiable sites remained in the denominator, which depresses the Ecuadorian percentages specifically, since all such cases occur in that group.

Vantage-point artefacts. The three non-Ecuadorian vantage points were reached through a commercial VPN and their egress addresses belong to data centres, so vendors that treat such addresses differently could in principle mimic a location effect. Three observations bound this concern: the mean number of WCAG violations per site, the rendering control, was identical across vantage points to within 1.1%; the German and United States endpoints share the same network type, so their contrast is free of the artefact; and one German endpoint was replaced by a second provider on a different autonomous system with an indistinguishable result. A symmetrical concern applies to the Ecuadorian endpoint, which is the only residential one: the exposure observed there might be a property of the particular access operator rather than of the country. This was tested directly. In September 2026 the full protocol was run over the 126 sites from a second Ecuadorian residential operator, Telconet (AS27947, Guayaquil), and repeated from the original operator, with three passes consolidated by the same majority rule in each series. In the same month, 84 of the 126 sites (66.7%) set at least one tracking cookie before consent under the second operator against 86 (68.3%) under the first; 122 sites are concordant and 4 discordant, with an exact McNemar p of 0.625. The comparison changes operator, city and machine together and cannot separate them, so it is informative only because no difference appears. Within that limit, the Ecuadorian result is not a property of a single access network. The residual fall in non-vendor cookies at the European and British vantage points, reported in Section 4.7, is not fully explained and is the weakest point of this part of

the design. The four-vantage replication and the five-vantage network campaign were run on consecutive days, 15 and 16 August, so the vendor behaviour they describe is that of those two days. The network records attribute a request to a vendor by its domain, so a vendor serving code from a domain outside the deposited list would be missed, and a request withheld upstream of a vendor cannot be assigned to the site or to another third party with these data. A further limitation concerns which sites the replication could retain. Five of the 126 failed to respond on at least one of the twelve passes and are excluded from the paired analysis, and four of those five are Ecuadorian; the loss is therefore not random with respect to group, and it points in the direction of lower availability among the Ecuadorian portals. Because the sensitivity analysis reported in Section 4.7 leaves the conclusion unchanged, the effect on the vantage-point comparison is negligible, but the differential availability itself is worth recording as an observation about the infrastructure of the two groups.

Measurement stability. The documentary review remains a single observation. Pre-consent cookie state is not deterministic: it depends on A/B tests, ad auctions, content delivery network (CDN) node and deployment timing. For the live audit this was addressed by the three passes per vantage point, whose between-pass agreement ranged from 95.9% to 100%; for the group comparison of Section 4.2, which rests on the August single-pass measurement, it was not. Between occasions the deposit now bounds it for the tracking measurement: a second ETAPA series of three passes on 14 September 2026, holding operator, city, machine, instrument and software versions fixed, returns 68.0 % of sites tracking on each occasion, with two of 125 sites discordant (exact McNemar $p = 1.000$); the 126th site, UNEMI, returned no valid Ecuadorian pass in August and has no verdict to compare. That bounds the drift of the tracking verdict over a month. The same two series also ran axe-core, and the level-A verdict is the same on both occasions at 123 of 125 sites, 94 failing at each (exact McNemar $p = 1.000$); the single-pass accessibility measurement of Section 4.4 was not itself repeated. Only the home page was audited, whereas WCAG conformance is defined over pages and complete processes, and the transactional paths this article invokes as its motivation, admission and enrolment, were not measured at all. The banner and CMP detectors are heuristics whose false-negative rates are unknown.

Statistical conclusion validity. Three families of comparisons were corrected by the Holm procedure, the twelve indicators of Table 1, the six vantage-point pairs of Table 6 and the ten pairs of the five-vantage campaign, and only the adjusted values are used to support claims; the regional breakdown is reported without inference for the reason given in Section 4.1. The equivalence margins in the TOST analysis were not pre-specified. The node-level accessibility distribution is subject to pseudo-replication and compositional constraint, as stated in Section 4.4. Failure counts were not normalised by page complexity, so a large portal will accumulate more failing nodes than a small one without necessarily being less accessible.

External validity. The group comparison rests on a single pass from one country on 14 August 2026 and the replication on a single further day, 15 August, so the results describe the state of 126 home pages in August 2026. The composition of the benchmark group also depends on the three choices of the aggregation rule described in Section 3.2, which were not varied; it is one defensible selection among several. Web content changes; no claim in this article should be read as a durable property of any institution. One month later, under the same operator, city, machine and instrument, the pre-consent tracking verdict changed at two of 125 sites, and the level-A accessibility verdict at two of 125, which bounds that drift without making any of it durable.

6. Conclusions and future work

Measured against a benchmark of 63 top-ranked universities, the 63 Ecuadorian HEIs showed their widest and most robust gaps in privacy governance, above all in the identification of a privacy contact (38.1% compared with 85.7%) and in the publication of a first-level privacy notice (61.9% compared with 93.7%). In accessibility, both groups failed, but the failures were distributed differently: 45% of Ecuadorian failing nodes were at the basic level A compared with 19% in the benchmark group, an ordering preserved when the level-AAA rules are excluded. In pre-consent tracking, observed from Ecuador, the study found no difference between the groups and had no power to establish that they are equivalent. Within each group, measured accessibility and observed restraint in tracking were not associated.

Replicating the live audit from four vantage points then established that pre-consent exposure varies with the visitor's location rather than the institution's. On 15 August 2026 the same 121 pages set tracking cookies for 67.8% of visits from Ecuador and 60.3% of visits from Germany (Cochran Q, $p = 0.0012$; Ecuador–Germany Holm-adjusted $p = 0.023$ over the six pairwise comparisons), and network records from a second campaign the following day located the decision: at 13 of the 14 sites where Microsoft set cookies for a visitor in Ecuador, requests to Microsoft were still sent from Frankfurt and no cookie resulted, on Ecuadorian, United States, Canadian, Australian and Hong Kong university sites alike, whereas the request to TapAd was not sent from Europe at all.

These findings support a practical conclusion, a legal one and a distributive one. The practical conclusion is that Ecuador's distinctive shortfalls are in governance and in basic-level accessibility, and both are remediable without new legislation. The legal conclusion is that, for Microsoft and in part for Meta and LinkedIn, a duty European law places on the university is in practice being discharged by advertising vendors that are themselves controllers, on the basis of unpublished country lists, which satisfies the visitor in Frankfurt while leaving neither party demonstrably accountable: the university cannot account for conduct it does not observe, and the vendor's own obligations are not being supervised. The distributive conclusion is the one that returns this article to its starting point: the applicants who receive no protection are those outside the perimeter those lists describe, which in this measurement included applicants browsing from Ecuador, and their exposure is invisible to anyone testing from inside it.

Four further items follow from the threats stated in Section 5.6: independent double coding by human coders, without assistance, of at least 30% of the sample, with a Cohen's κ per indicator and full reconciliation of discrepancies; extension of the audit to at least five pages per site, including an admission or enrolment path, with the page as the unit and the site as a random effect; more robust tracking measures than the cookie, in particular the count of distinct third-party domains contacted before consent; and an internal analysis of the Ecuadorian group by institution type, size and platform, for which the necessary variable is already recorded in Appendix D and which is likely to be of greater local value than any international comparison. To these the present results add a fifth: mapping the country lists operated by the major tag vendors, by measuring from one vantage point per candidate jurisdiction, so that the perimeter within which protection is delivered can be described empirically rather than assumed to coincide with any legal boundary.

Supplementary information

The audit script, its configuration, the raw per-site JavaScript Object Notation (JSON) output of the twelve multi-vantage passes together with the location-verification records, the complete cookie inventory with its classification under both taxonomies, the documentary matrix, the codebook and the analysis scripts are deposited in a public repository (Guerrero-Ulloa et al., 2026). Every figure reported in Section 4.7, including the vendor-level attribution and the network-layer analysis of Table 7, is reproducible from those files with the analysis scripts released alongside them.

Data availability statement

The datasets generated and analysed during the current study are openly available in the Zenodo repository (Guerrero-Ulloa et al., 2026), under the concept Digital Object Identifier `10.5281/zenodo.22405998`, which resolves to the latest version. The deposit is licensed CC BY 4.0 for data and documentation and MIT for code.

Code availability

The audit scripts and the analysis scripts are available in the same repository (Guerrero-Ulloa et al., 2026), under the MIT licence. The group comparisons of Table 1 use standard routines (Fisher exact test, Newcombe and Woolf intervals, Holm adjustment) applied to the per-site counts in the deposited matrix.

Use of generative artificial intelligence

Every decision reported in this article was taken by the authors, who operated every measurement instrument and are accountable for every analysis and for the final version of the text. A generative artificial intelligence assistant, Claude (Anthropic), used through its web and desktop applications between August and November 2026, served as a tool under that accountability: to review and correct prose and auxiliary analysis scripts of the manuscript and the deposit, and to draft some of those passages and scripts, among them the network-layer analysis of Section 4.7, reviewed afterwards by the first author; and to recompute reported quantities from the deposited data as a check on the manuscript. It was not used to operate the audit instrument or the network campaigns, or to decide what the study concludes, and it generated no live measurement; the instrument and the software versions are those given in Sections 3.5, 3.6 and 3.9. It does not satisfy the authorship criteria of this journal and is not credited as an author.

Two uses of it reached the data, and are recorded as such. First, the documentary matrix of Section 3.4 was compiled with its assistance: it retrieved each institution's documents and proposed the code for each cell, the three authors and one external collaborator took part in coding and reviewing the result, and six institutions were then re-verified by hand, cell by cell, which corrected two cells; the individual codings were not retained. Second, of the 349 image elements that rule `qt1vmo` refers to human judgement in the first coding of the manual validation, 345 were resolved with its assistance and reviewed by the first author, and the remaining four by the first author

alone. No verdict reported in this article rests on those resolutions. The site verdicts of Section 4.5 are those of the second, wholly human coding described in Section 3.6, the assisted resolutions appear only as a sensitivity reading, and they are excluded from every agreement estimate.

Disclosure statement

The corresponding author and the third author are affiliated with Universidad Técnica Estatal de Quevedo, which is one of the 63 Ecuadorian institutions assessed in this study and appears in Appendix D. The second author is affiliated with the Ministry of Education of Ecuador, a public body of the same national education system. The authors participated in this study as individual researchers and not on behalf of either institution. Neither institution proposed, commissioned, funded or approved the work, and neither had any role in its design, in the collection or coding of data, in the analysis or in the decision to publish. The coding of the authors' own institution follows the same codebook as every other site and is reported in Appendix D without exception.

Funding

No funding was received for conducting this study.

Author contributions

G.G.-U.: conceptualisation, methodology, software, formal analysis, investigation, visualisation, resources, data curation, project administration, supervision, writing – original draft, writing – review and editing. A.Z.P.: investigation, data curation, validation, writing – review and editing. E.D.-M.: investigation, software, validation, writing – review and editing. All authors read and approved the final manuscript. No funding acquisition role is declared, as the study received no funding.

Ethics approval and consent to participate

Not applicable. The study analysed publicly accessible web pages and involved no human participants and no animals. No personal data were collected: the privacy-contact indicator records only whether an institution identifies a privacy contact, and no contact detail is reproduced in the article or in the deposit.

Consent for publication

Not applicable.

References

Abascal, J., Barbosa, S. D. J., Nicolle, C., & Zaphiris, P. (2016). Rethinking universal accessibility: A broader approach considering the digital gap. *Universal Access in the Information Society*, *15*(2), 179–182. https://doi.org/10.1007/s10209-015-0416-1

Acosta-Vargas, P., González, M., & Luján-Mora, S. (2020). Dataset for evaluating the accessibility of the websites of selected Latin American universities. *Data in Brief*, *28*, 105013. https://doi.org/10.1016/j.dib.2019.105013

Acosta-Vargas, P., Ramos-Galarza, C., Salvador-Ullauri, L., Chanchi, G. E., & Jadán-Guerrero, J. (2020). Improve accessibility and visibility of selected university websites. In I. L. Nunes (Ed.), *Advances in human factors and systems interaction (AHFE 2020)* (Vol. 1207, pp. 229–235). Springer. https://doi.org/10.1007/978-3-030-51369-6_31

Aljedaani, W. (2025). Evaluating the accessibility of E-government websites in the United States: Empirical evidence and insights. *Universal Access in the Information Society*, *24*(3), 2845–2866. https://doi.org/10.1007/s10209-025-01229-z

Asamblea Nacional del Ecuador. (2021). *Ley Orgánica de Protección de Datos Personales.* Registro Oficial No. 459, Quinto Suplemento, 26 May 2021.

Belcheva, V., Ermakova, T., & Fabian, B. (2023). Understanding website privacy policies—a longitudinal analysis using natural language processing. *Information*, *14*(11), 622. https://doi.org/10.3390/info14110622

Bollinger, D., Kubicek, K., Cotrini, C., & Basin, D. (2022). Automating cookie consent and GDPR violation detection. In *31st USENIX security symposium (USENIX security 22)* (pp. 2893–2910). USENIX Association. `https://www.usenix.org/conference/usenixsecurity22/presentation/bollinger`

California State Legislature. (2018). *California Consumer Privacy Act of 2018.* Cal. Civ. Code §1798.100 et seq.; amended by the California Privacy Rights Act of 2020.

Campoverde-Molina, M., Luján-Mora, S., & Valverde, L. (2023). Accessibility of university websites worldwide: A systematic literature review. *Universal Access in the Information Society*, *22*(1), 133–168. https://doi.org/10.1007/s10209-021-00825-z

Cao, S., & Loiacono, E. (2022). Perceptions of web accessibility guidelines by student website and app developers. *Behaviour & Information Technology*, *41*(12), 2616–2634. https://doi.org/10.1080/0144929X.2021.1940278

Carrillo, A. J., & Jackson, M. (2022). Follow the leader? A comparative law study of the EU's General Data Protection Regulation's impact in Latin America. *ICL Journal*, *16*(2), 177–262. https://doi.org/10.1515/icl-2021-0037

Chavarría-Mora, E. (2025). (Lack of) patterns in commitment: Data protection in the Latin America and Caribbean personal data protection laws. *Social Media + Society*, *11*(2), 20563051251337206. https://doi.org/10.1177/20563051251337206

Court of Justice of the European Union. (2019, July). *Judgment of 29 July 2019, Fashion ID GmbH & Co. KG v Verbraucherzentrale NRW eV, Case C-40/17.* ECLI:EU:C:2019:629. `https://curia.europa.eu/juris/liste.jsf?num=C-40/17` (Decided under Directive 95/46/EC)

Dabrowski, A., Merzdovnik, G., Ullrich, J., Sendera, G., & Weippl, E. (2019). Measuring cookies and web privacy in a post-GDPR world. In D. Choffnes & M. Barcellos (Eds.), *Passive and active measurement (PAM 2019)* (Vol. 11419, pp. 258–270). Springer. https://doi.org/10.1007/978-3-030-15986-3_17

Degeling, M., Utz, C., Lentzsch, C., Hosseini, H., Schaub, F., & Holz, T. (2019). We value your privacy . . . now take some cookies: Measuring the GDPR's impact on web privacy. In *Proceedings of the 2019 network and distributed system security symposium (NDSS 2019).* Internet Society. https://doi.org/10.14722/ndss.2019.23378

Deque Systems. (2025). *axe-core: Accessibility engine for automated Web UI testing, version 4.13.* `https://github.com/dequelabs/axe-core`. (Accessed 11 August 2026)

Englehardt, S., & Narayanan, A. (2016). Online tracking: A 1-million-site measurement and analysis. In *Proceedings of the 2016 ACM SIGSAC conference on computer and com-*

munications security (CCS '16) (pp. 1388–1401). Association for Computing Machinery. https://doi.org/10.1145/2976749.2978313

Epstein, D., & Quinn, K. (2025). A dimensionalised privacy behaviour model: An empirical test of a conceptual proposition. *Behaviour & Information Technology*, *44*(15), 3621–3641. https://doi.org/10.1080/0144929X.2024.2443523

Etxabe-Antia, A., Beitia-Amondarain, A., González de Heredia-López de Sabando, A., & Justel-Lozano, D. (2025). Characterisation of accessibility guidelines for digital technologies. *Universal Access in the Information Society*, *24*(3), 2105–2125. https://doi.org/10.1007/s10209-025-01214-6

European Data Protection Board. (2019, November). *Guidelines 3/2018 on the territorial scope of the GDPR (Article 3)* (Guidelines, version adopted after public consultation). Author. `https://www.edpb.europa.eu/our-work-tools/our-documents/guidelines/guidelines-32018-territorial-scope-gdpr-article-3-version_en`

European Data Protection Board. (2021, July). *Guidelines 07/2020 on the concepts of controller and processor in the GDPR* (Guidelines, Version 2.1). Author. `https://www.edpb.europa.eu/system/files/2021-07/eppb_guidelines_202007_controllerprocessor_final_en.pdf` (Version 2.0 adopted 7 July 2021; version 2.1, minor corrections, 20 September 2022)

European Data Protection Board. (2024, October). *Guidelines 2/2023 on technical scope of art. 5(3) of the ePrivacy Directive* (Guidelines, Version 2.0). Author. `https://www.edpb.europa.eu/system/files/2024-10/edpb_guidelines_202302_technical_scope_art_53_eprivacydirective_v2_en_0.pdf`

European Parliament and Council of the European Union. (2002). *Directive 2002/58/EC Concerning the Processing of Personal Data and the Protection of Privacy in the Electronic Communications Sector (ePrivacy Directive).* Official Journal of the European Union, L 201, 31 July 2002, pp. 37–47; amended by Directive 2009/136/EC.

European Parliament and Council of the European Union. (2016). *Regulation (EU) 2016/679 on the Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data (General Data Protection Regulation).* Official Journal of the European Union, L 119, 4 May 2016, pp. 1–88.

European Parliament and Council of the European Union. (2019). *Directive (EU) 2019/882 on the Accessibility Requirements for Products and Services (European Accessibility Act).* Official Journal of the European Union, L 151, 7 June 2019, pp. 70–115.

Fabian, B., Ermakova, T., & Lentz, T. (2017). Large-scale readability analysis of privacy policies. In *Proceedings of the international conference on web intelligence (WI '17)* (pp. 18–25). Association for Computing Machinery. https://doi.org/10.1145/3106426.3106427

Fakrudeen, M. (2025). Evaluation of the accessibility and usability of university websites: A comparative study of the Gulf region. *Universal Access in the Information Society*, *24*(2), 1883–1898. https://doi.org/10.1007/s10209-024-01160-9

Fiers, W., Boland Jr., K., Doyle, E., Kraft, M. J., & Kirkpatrick, A. (2026, February). *Accessibility Conformance Testing (ACT) Rules Format 1.1* (W3C Recommendation). World Wide Web Consortium. `https://www.w3.org/TR/act-rules-format/` (Recommendation of 5 February 2026)

Fischer, T., Lundell, B., & Gamalielsson, J. (2025). Coverage of web accessibility guidelines provided by automated checking tools. *Universal Access in the Information Society*, *24*(4), 3615–3637. https://doi.org/10.1007/s10209-025-01263-x

Government of Japan. (2003). *Act on the Protection of Personal Information.* Act No. 57 of 2003, as amended.

Gray, C. M., Santos, C., Bielova, N., Toth, M., & Clifford, D. (2021). Dark patterns and the legal requirements of consent banners: An interaction criticism perspective. In *Proceedings of the 2021 CHI conference on human factors in computing systems (CHI '21)* (pp. 1–18). Association for Computing Machinery. https://doi.org/10.1145/3411764.3445779

Guerrero-Ulloa, G., Zuñiga Paredes, A., & Díaz-Macías, E. (2026). *WCAG conformance and pre-consent tracking on Ecuadorian and top-ranked university websites: audit dataset,*

multi-vantage measurements and analysis code [Data set]. Zenodo. (Dataset. Concept DOI, resolving to the latest version. Data and documentation under CC BY 4.0; code under MIT) https://doi.org/10.5281/zenodo.22405998

Habib, H., Li, M., Young, E., & Cranor, L. (2022). "Okay, whatever": An evaluation of cookie consent interfaces. In *Proceedings of the 2022 CHI conference on human factors in computing systems (CHI '22)* (pp. 1–27). Association for Computing Machinery. https://doi.org/10.1145/3491102.3501985

Hermosa-Ramírez, I. (2025). Universal and ecological design in media accessibility: Finding common ground. *Universal Access in the Information Society*, *24*(3), 1977–1984. https://doi.org/10.1007/s10209-023-01077-9

Iniesto, F., & Rodrigo, C. (2025). Service-learning for accessibility: Understanding WCAG errors and successes in Spanish city council websites. *Universal Access in the Information Society*, *24*(4), 3241–3255. https://doi.org/10.1007/s10209-025-01252-0

International Organization for Standardization. (2025). *ISO/IEC 40500:2025 — information technology — W3C Web Content Accessibility Guidelines (WCAG) 2.2* (International Standard). International Organization for Standardization. `https://www.iso.org/standard/58625.html` (Approved 21 October 2025; available free of charge)

Krejtz, K., Marcus-Quinn, A., Duarte, C., Stasiak, I., Seixas Pereira, L., & Krejtz, I. (2025). Higher education accessibility information in practice: A report on the accessibility of European universities. *Universal Access in the Information Society*, *24*(3), 2673–2685. https://doi.org/10.1007/s10209-025-01224-4

Kretschmer, M., Pennekamp, J., & Wehrle, K. (2021). Cookie banners and privacy policies: Measuring the impact of the GDPR on the web. *ACM Transactions on the Web*, *15*(4), 1–42. https://doi.org/10.1145/3466722

Krittika, K., & Shimray, S. R. (2025). Accessibility evaluation of top 25 QS-ranked world universities and top 25 QS-ranked Indian universities using WCAG 2.2: A comparative study. *Universal Access in the Information Society*, *24*(3), 2889–2902. https://doi.org/10.1007/s10209-025-01218-2

Legislative Assembly of British Columbia. (1996). *Freedom of Information and Protection of Privacy Act.* RSBC 1996, c. 165.

Legislative Assembly of Ontario. (1990). *Freedom of Information and Protection of Privacy Act.* R.S.O. 1990, c. F.31.

Legislative Council of Hong Kong. (1995). *Personal Data (Privacy) Ordinance.* Cap. 486.

Li, Y., Rho, E. H. R., & Kobsa, A. (2022). Cultural differences in the effects of contextual factors and privacy concerns on users' privacy decision on social networking sites. *Behaviour & Information Technology*, *41*(3), 655–677. https://doi.org/10.1080/0144929X.2020.1831608

Libert, T. (2015). Exposing the hidden web: An analysis of third-party HTTP requests on 1 million websites. *International Journal of Communication*, *9*, 3544–3561. `https://ijoc.org/index.php/ijoc/article/view/3646`

Lindegren, D., Karegar, F., Kane, B., & Pettersson, J. S. (2021). An evaluation of three designs to engage users when providing their consent on smartphones. *Behaviour & Information Technology*, *40*(4), 398–414. https://doi.org/10.1080/0144929X.2019.1697898

Liu, Q., & Khalil, M. (2023). Understanding privacy and data protection issues in learning analytics using a systematic review. *British Journal of Educational Technology*, *54*(6), 1715–1747. https://doi.org/10.1111/bjet.13388

Magrizos, S., Campora, M., Lamprinakos, G., Giovanis, A., & Christofi, M. (2026). Transparency by design: The effect of privacy policies visualisation on brand trust and perceived intrusion. *Behaviour & Information Technology*, *45*(3), 450–462. https://doi.org/10.1080/0144929X.2025.2520594

Maldonado-Garcés, V., Sánchez-García, J. C., Hernández-Sánchez, B., Acosta-Vargas, P., & Araujo, E. (2025). Physical accessibility in higher education: Evaluating a university campus in Ecuador for sustainable inclusion. *Sustainability*, *17*(12), 5652. https://doi.org/10.3390/su17125652

Matte, C., Bielova, N., & Santos, C. (2020). Do cookie banners respect my choice? Measur-

ing legal compliance of banners from IAB Europe's Transparency and Consent Framework. In *2020 IEEE symposium on security and privacy (SP)* (pp. 791–809). IEEE. https://doi.org/10.1109/SP40000.2020.00076

Meier, Y., & Krämer, N. C. (2025). Differences in access to privacy information can partly explain digital inequalities in privacy literacy and self-efficacy. *Behaviour & Information Technology*, *44*(6), 1183–1198. https://doi.org/10.1080/0144929X.2024.2349183

Microsoft. (2026). *Playwright: Reliable end-to-end testing for modern web apps.* `https://playwright.dev`. (Accessed 11 August 2026)

Munot, S., Fiebig, T., & Kaur, M. (2025). Compliance by design or by disguise? GDPR's reshaping of universities' privacy policies. In *Proceedings of the 24th workshop on privacy in the electronic society (WPES '25)* (pp. 183–197). Association for Computing Machinery. https://doi.org/10.1145/3733802.3764047

Mutimukwe, C., Viberg, O., McGrath, C., & Cerratto-Pargman, T. (2026). Privacy in online proctoring systems in higher education: Stakeholders' perceptions, awareness and responsibility. *Journal of Computing in Higher Education*, *38*(2), 732–761. https://doi.org/10.1007/s12528-025-09461-5

National Assembly of Quebec. (1982). *Act Respecting Access to Documents Held by Public Bodies and the Protection of Personal Information.* CQLR c. A-2.1; amended by S.Q. 2021, c. 25.

National Assembly of the Republic of Korea. (2011). *Personal Information Protection Act.* Act No. 10465.

National People's Congress of the People's Republic of China. (2021). *Personal Information Protection Law.* Effective 1 November 2021.

Nouwens, M., Liccardi, I., Veale, M., Karger, D., & Kagal, L. (2020). Dark patterns after the GDPR: Scraping consent pop-ups and demonstrating their influence. In *Proceedings of the 2020 CHI conference on human factors in computing systems (CHI '20)* (pp. 1–13). Association for Computing Machinery. https://doi.org/10.1145/3313831.3376321

Parliament of Australia. (1988). *Privacy Act 1988 (Cth) and the Australian Privacy Principles.* No. 119 of 1988.

Parliament of Canada. (2000). *Personal Information Protection and Electronic Documents Act.* S.C. 2000, c. 5.

Parliament of Singapore. (2012). *Personal Data Protection Act 2012.* Act No. 26 of 2012.

Parliament of the United Kingdom. (2018). *Data Protection Act 2018.* 2018 c. 12.

Persson, H., Åhman, H., Yngling, A. A., & Gulliksen, J. (2015). Universal design, inclusive design, accessible design, design for all: Different concepts—one goal? On the concept of accessibility—historical, methodological and philosophical aspects. *Universal Access in the Information Society*, *14*(4), 505–526. https://doi.org/10.1007/s10209-014-0358-z

Presidencia de la República del Ecuador. (2023). *Reglamento General a la Ley Orgánica de Protección de Datos Personales.* Executive Decree No. 904; Registro Oficial Supplement No. 435, 13 November 2023.

QS Quacquarelli Symonds. (2025). *QS World University Rankings 2026.* `https://www.topuniversities.com/world-university-rankings/2026`. (Accessed 11 August 2026)

Rasaii, A., Singh, S., Gosain, D., & Gasser, O. (2023). Exploring the cookieverse: A multi-perspective analysis of web cookies. In A. Brunstrom, M. Flores, & M. Fiore (Eds.), *Passive and active measurement (PAM 2023)* (Vol. 13882, pp. 623–651). Springer. https://doi.org/10.1007/978-3-031-28486-1_26

Sanchez-Rola, I., Dell'Amico, M., Kotzias, P., Balzarotti, D., Bilge, L., Vervier, P.-A., & Santos, I. (2019). Can I opt out yet? GDPR and the global illusion of cookie control. In *Proceedings of the 2019 ACM Asia conference on computer and communications security (AsiaCCS '19)* (pp. 340–351). Association for Computing Machinery. https://doi.org/10.1145/3321705.3329806

ShanghaiRanking Consultancy. (2025). *Academic Ranking of World Universities 2025.* `https://www.shanghairanking.com/rankings/arwu/2025`. (Accessed 11 August 2026)

Spyridonis, F., & Daylamani-Zad, D. (2021). A serious game to improve engagement

with web accessibility guidelines. *Behaviour & Information Technology*, *40*(6), 578–596. https://doi.org/10.1080/0144929X.2019.1711453

Swiss Federal Assembly. (2020). *Federal Act on Data Protection.* SR 235.1, of 25 September 2020; in force since 1 September 2023.

Times Higher Education. (2025). *World University Rankings 2026.* `https://www.timeshighereducation.com/world-university-rankings/latest/world-ranking`. (Accessed 11 August 2026)

Trevisan, M., Traverso, S., Bassi, E., & Mellia, M. (2019). 4 years of EU cookie law: Results and lessons learned. *Proceedings on Privacy Enhancing Technologies*, *2019*(2), 126–145. https://doi.org/10.2478/popets-2019-0023

United States Congress. (1974). *Family Educational Rights and Privacy Act.* 20 U.S.C. §1232g.

United States Congress. (1986). *Section 508 of the Rehabilitation Act of 1973, Electronic and Information Technology.* 29 U.S.C. §794d.

United States Congress. (1990). *Americans with Disabilities Act of 1990.* Pub. L. No. 101–336, 104 Stat. 327; 42 U.S.C. §12101 et seq.

United States Congress. (1996). *Health Insurance Portability and Accountability Act.* Pub. L. No. 104–191, 110 Stat. 1936.

van Eijk, R., Asghari, H., Winter, P., & Narayanan, A. (2019). The impact of user location on cookie notices (inside and outside of the European Union). In *Workshop on technology and consumer protection (ConPro '19), co-located with the 40th IEEE symposium on security and privacy.* `https://www.ieee-security.org/TC/SPW2019/ConPro/papers/vaneijk-conpro19.pdf`

W3C Web Accessibility Initiative. (2026). *ACT rules* (Accessibility Conformance Testing rule catalogue). World Wide Web Consortium. `https://www.w3.org/WAI/standards-guidelines/act/rules/` (Rules `23a2a8`, `qt1vmo`, `e88epe`, `afw4f7`, `c487ae`, `5effbb` and `fd3a94`)

World Wide Web Consortium. (2008, December). *Web Content Accessibility Guidelines (WCAG) 2.0* (W3C Recommendation). Author. `https://www.w3.org/TR/2008/REC-WCAG20-20081211/`

World Wide Web Consortium. (2018, June). *Web Content Accessibility Guidelines (WCAG) 2.1* (W3C Recommendation). Author. `https://www.w3.org/TR/2018/REC-WCAG21-20180605/`

World Wide Web Consortium. (2024, December). *Web Content Accessibility Guidelines (WCAG) 2.2* (W3C Recommendation). Author. `https://www.w3.org/TR/2024/REC-WCAG22-20241212/`

Appendix A. Codebook and operational definitions

Table A1 gives the operational definition, the decision rule and the treatment of ambiguous cases for each indicator. The tracking-cookie taxonomy follows.

Table A1. Operational definitions of the coded indicators.

Indicator	Coded present when	Coded absent or partial when
Privacy notice	A document describing the processing of personal data is reachable from the home page or one click below it	**P** if its stated scope covers only part of the institution's processing; absent if unreachable from the home page

Continued on next page

(Table A1, continued)

Indicator	Coded present when	Coded absent or partial when
Framework cited	The notice names a specific instrument, for example the LOPDP, the GDPR or the Personal Data (Privacy) Ordinance	Absent for generic formulations such as "applicable law"; **P** for a named framework that does not apply to the institution
Rights listed	At least three of access, rectification, erasure, restriction, objection and portability are enumerated, with a route to exercise them	Absent for an unenumerated mention of "your rights"; **P** if enumerated without any route
Privacy contact or DPO	A named role, a dedicated address or a dedicated form for privacy matters is given	Absent for a general institutional contact form or switchboard number
Dedicated cookie policy	A document specifically about cookies or similar tracking technologies is published and reachable under the same rule as the privacy notice	Absent when cookies are covered only as a section inside a general privacy notice. Benchmark group only
Accessibility declared	An accessibility statement, an accessibility policy or an accessibility toolbar is published	Absent when accessibility is mentioned only inside another document
Transport security	The home page is served over HTTPS with a chain that a current browser accepts without warning	Absent for expired, self-signed or incomplete chains
Tracking before consent	At least one cookie whose name matches the taxonomy below is present before any interaction with a consent interface	Absent when no matching cookie is present
No level-A failure	axe-core reports no violation carrying a level-A tag	Absent when at least one level-A violation is reported
1.1.1 non-text content (manual)	Every applicable `img`, `canvas` or `svg` element has a non-empty accessible name (ACT `23a2a8`) that serves a purpose equivalent to the non-text content (ACT `qt1vmo`); an element outside the accessibility tree is treated as decorative (ACT `e88epe`)	Failed when any applicable element has an empty accessible name or a name that does not serve an equivalent purpose. Programmatically hidden elements, and elements with an ancestor named from author, are inapplicable
1.4.3 contrast, minimum (manual)	Every visible character in a text node reaches a contrast ratio of 4.5:1, or 3.0:1 for large-scale text (ACT `afw4f7`)	Failed when any applicable character falls below the threshold. Text with a disabled ancestor, text used in the accessible name of a disabled widget, purely decorative text and text not expressing anything in human language are inapplicable; visually hidden text is not a visible character and is therefore outside the rule

Continued on next page

(Table A1, continued)

Indicator	Coded present when	Coded absent or partial when
2.4.4 link purpose in context (manual)	Every link in the accessibility tree has a non-empty accessible name (ACT `c487ae`) which, together with its programmatically determined link context, describes the purpose of the link (ACT `5effbb`); links sharing an identical accessible name in the same context serve an equivalent purpose (ACT `fd3a94`)	Failed when a link has an empty accessible name, or when name and programmatically determined context together do not describe its purpose. A generic wrapping element is not a programmatically determined context

Tracking-cookie taxonomy. A cookie was classified as belonging to an advertising or analytics vendor when its name matched one of the following patterns, applied as regular expressions to the cookie name. The list was extended after the multi-vantage replication of Section 3.7; the families added at that point are marked with an asterisk. The extended list is applied throughout by reclassifying the stored cookie names, so that no measurement was repeated and every reported figure derives from the same recorded observations. At the site level the two lists differ on these data by a single site, an Ecuadorian institution whose only matching cookies belong to the Sourcebuster family. At the level of individual cookies they differ substantially, which is why the extension was made.

- *Google Analytics and Ads*: `^_ga`, `^_gid$`, `^__utm`, `^_gcl_`, `^_gac_`; *Privacy Sandbox**: `^receive-cookie-deprecation$`.
- *YouTube and DoubleClick*: `^YSC$`, `^VISITOR_INFO`, `^VISITOR_PRIVACY*`, `^__Secure-YNID$*`, `^__Secure-ROLLOUT*`, `^IDE$`, `^test_cookie$`, `^DSID$*`.
- *Meta*: `^_fbp$`, `^_fbc$`, `^fr$`. *TikTok*: `^_tt_`, `^_ttp$`, `^ttcsid*`.
- *Microsoft Clarity, Advertising and UET*: `^_clck$`, `^_clsk$`, `^MUID$`, `^MR$*`, `^SM$*`, `^SRM_B$*`, `^ANONCHK$*`, `^CLID$*`, `^_uet`.
- *LinkedIn Insight Tag**: `^bcookie$`, `^bscookie$`, `^lidc$`, `^li_sugr$`, `^li_gc$`, `^UserMatchHistory$`, `^AnalyticsSyncHistory$`.
- *TapAd**: `^TapAd_`. *StackAdapt**: `^sa-user-id`. *Snapchat**: `^_scid`, `^sc_at$`.
- *X and Twitter*: `^muc_ads$*`, `^personalization_id$`, `^_twpid$*`.
- *Hotjar*: `^_hj`. *Siteimprove*: `^nmstat$`. *Matomo*: `^_pk_`. *PixelYourSite and Sourcebuster*: `^pys`, `^sbjs_*`. *Pinterest*: `^_pin_`.
- *Adobe and Tealium*: `^AMCV_`, `^s_`, `^utag`, `^mbox`; *ContentSquare**: `^_cs_`.
- *Baidu*: `^Hm_lvt_`, `^Hm_lpvt_`, `^HMACCOUNT`, `^BAIDUID$`. *Quantcast*: `^__qca$`. *LiveIntent*: `^_lc2_`, `^ln_or$`.

The list is name-based and therefore conservative: a vendor using an unlisted or randomised cookie name is not counted, which is one reason the figures in Section 4.2 are lower bounds. The consequence of the initial omissions is quantified in Section 4.7: under the original list the fall in vendor cookies at the European vantage point is 33.4%, and under the extended list it is 43.6%.

Appendix B. Indicator-by-jurisdiction mapping

Table B1 states, for each jurisdiction in the study, the instrument accepted as the applicable framework and whether the jurisdiction imposes a general duty of prior consent

for non-essential cookies. It is the basis for the distinction, maintained throughout Section 4, between observed conduct and legal non-compliance. Figure E2 in Appendix E places these regimes on a common timeline.

Table B1. Applicable framework and prior-consent duty by jurisdiction.

Jurisdiction	**Instrument accepted as applicable framework**	**General prior-consent duty for non-essential cookies**
Ecuador	LOPDP and its implementing regulation (Asamblea Nacional del Ecuador, 2021; Presidencia de la República del Ecuador, 2023)	No equivalent of Art. 5(3) ePrivacy; consent is required for processing, not as a storage-access rule
European Union	GDPR (European Parliament and Council of the European Union, 2016) and national transpositions of the ePrivacy Directive (European Parliament and Council of the European Union, 2002), which differ between the five member states represented in the benchmark group	Yes, under Art. 5(3) ePrivacy as transposed (European Data Protection Board, 2024)
United Kingdom	United Kingdom GDPR and Data Protection Act 2018 (Parliament of the United Kingdom, 2018)	Yes, under the national ePrivacy transposition
Switzerland	Federal Act on Data Protection (FADP) (Swiss Federal Assembly, 2020)	No general prior-consent rule; transparency and opt-out apply
United States	California Consumer Privacy Act (CCPA) (California State Legislature, 2018) for California; the Family Educational Rights and Privacy Act (FERPA) (United States Congress, 1974) and the Health Insurance Portability and Accountability Act (HIPAA) (United States Congress, 1996) govern education records and protected health information respectively and do not govern the public portal	No general prior-consent duty; opt-out model
Canada	Provincial public-sector statutes for public universities (Legislative Assembly of British Columbia, 1996; Legislative Assembly of Ontario, 1990; National Assembly of Quebec, 1982); the Personal Information Protection and Electronic Documents Act (PIPEDA) (Parliament of Canada, 2000) for commercial activity	No general prior-consent duty; implied consent recognised
China	Personal Information Protection Law (PIPL) (National People's Congress of the People's Republic of China, 2021)	Consent required for non-essential processing
Japan	Act on the Protection of Personal Information (APPI) (Government of Japan, 2003)	No general prior-consent duty for cookies as such
South Korea	Personal Information Protection Act (PIPA) (National Assembly of the Republic of Korea, 2011)	Consent required for non-essential processing
Singapore	Personal Data Protection Act (PDPA) (Parliament of Singapore, 2012)	Deemed consent model; no general prior-consent duty

Continued on next page

(Table B1, continued)

Jurisdiction	Instrument accepted as applicable framework	General prior-consent duty for non-essential cookies
Australia	Privacy Act 1988 (Parliament of Australia, 1988)	No general prior-consent duty
Hong Kong SAR	Personal Data (Privacy) Ordinance (PDPO) (Legislative Council of Hong Kong, 1995)	No general prior-consent duty

The consequence for interpretation is direct: in the majority of the jurisdictions studied, setting a tracking cookie before consent is not in itself unlawful. The indicator is retained because it measures the exposure of the visitor, which is the concern of this article, but it is not, and is nowhere presented as, a compliance rate.

Appendix C. Documentary matrix, benchmark group

Legend: ✓ present; – absent; **P** partial, meaning limited scope or a generic framework; *NV* not verifiable by this method. Rankings: QS 2026, THE 2026 and ARWU 2025, giving the position, with "–" for absence from that ranking's top 75. Country in ISO 3166-1 alpha-2. Columns: Not. (first-level privacy notice), Frm. (specific legal instrument named), Rts. (data-subject rights enumerated), DPO (privacy contact or data protection officer identified), Ckp. (a dedicated cookie policy is published, a documentary indicator distinct from the live banner measurement of Section 4.2 and reported in aggregate in Section 4.1), Acc. (accessibility statement or tools declared). Coding date: August 2026. Cells marked † were corrected after the manual re-verification described in Section 3.4.

Table C1. Documentary matrix of the 63 benchmark institutions.

#	Abbr.	C	QS	THE	ARWU	Not.	Frm.	Rts.	DPO	Ckp.	Acc.
1	MIT	US	1	2	3	✓	✓	✓	✓	–	✓
2	Stanford	US	3	5	2	✓	✓	✓	✓	✓	✓
3	Harvard	US	5	5	1	✓	–†	–	–	–	✓
4	Oxford	GB	4	1	6	✓	✓	✓	✓	✓	✓
5	Cambridge	GB	6	3	4	✓†	**P**	–	–	✓	✓
6	Caltech	US	10	7	9	✓	✓	–	✓	–	✓
7	UC Berkeley	US	17	9	5	✓	✓	–	✓	–	✓
8	Princeton	US	25	3	7	✓	✓	✓	✓	–	✓
9	Imperial	GB	2	8	26	✓	✓	**P**	–	✓	✓
10	U Chicago	US	13	15	10	✓	**P**	✓	✓	–	–
11	ETH Zurich	CH	7	11	22	✓	✓	✓	✓	–	✓
12	Yale	US	21	10	11	✓	✓	**P**	✓	–	✓
13	UPenn	US	15	14	14	✓	**P**	✓	✓	–	✓
14	UCL	GB	9	22	14	✓	**P**	✓	✓	✓	✓
15	Cornell	US	16	18	12	✓	**P**	✓	✓	–	✓
16	Tsinghua	CN	17	12	18	✓	✓	✓	✓	–	–
17	Peking	CN	14	13	23	**P**	**P**	–	–	–	–
18	Johns Hopkins	US	24	16	19	✓	✓	✓	✓	–	✓
19	Columbia	US	38	20	8	✓	✓	✓	✓	✓	✓
20	Toronto	CA	29	21	25	✓	✓	✓	✓	–	✓
21	UCLA	US	46	18	16	✓	✓	✓	✓	–	✓
22	NUS	SG	8	17	56	✓	✓	✓	✓	–	–
23	U Tokyo	JP	36	26	31	✓	✓	✓	–	–	–

Continued on next page

(Table C1, continued)

#	Abbr.	C	QS	THE	ARWU	Not.	Frm.	Rts.	DPO	Ckp.	Acc.
24	TUM	DE	22	27	45	✓	✓	✓	✓	–	–
25	Melbourne	AU	19	37	38	✓	✓	✓	✓	–	–
26	Edinburgh	GB	34	29	37	✓	✓	✓	✓	–	✓
27	EPFL	CH	22	35	44	✓	✓	✓	✓	–	✓
28	Michigan	US	45	23	33	✓	–	–	✓	–	–
29	Northwestern	US	42	30	31	✓	**P**	✓	✓	✓	✓
30	Fudan	CN	30	36	41	–	–	–	–	–	–
31	PSL	FR	28	48	34	✓	✓	✓	✓	–	–
32	HKU	HK	11	33	67	✓	✓	✓	✓	–	✓
33	Zhejiang	CN	49	39	24	✓	**P**	✓	✓	–	–
34	NYU	US	55	31	28	✓	✓	✓	✓	–	✓
35	SJTU	CN	47	40	30	–	–	–	–	–	–
36	KCL	GB	31	38	61	✓	**P**	✓	✓	✓	✓
37	UCSD	US	66	47	20	✓	✓	✓	✓	–	✓
38	LMU	DE	58	34	42	✓	✓	✓	✓	–	✓
39	Duke	US	62	28	46	✓	✓	✓	✓	–	✓
40	Manchester	GB	35	56	46	✓	✓	**P**	✓	–	✓
41	UBC	CA	40	45	53	✓	✓	**P**	✓	–	✓
42	Sydney	AU	25	53	72	✓	✓	✓	✓	–	✓
43	Paris-Saclay	FR	70	68	13	✓	✓	✓	✓	✓	✓
44	Kyoto	JP	57	61	46	✓	**P**	✓	✓	–	–
45	Illinois	US	70	41	53	✓	✓	✓	✓	–	✓
46	UT Austin	US	68	50	49	✓	✓	✓	✓	–	✓
47	UW	US	–	25	17	✓	✓	✓	✓	–	✓
48	NTU	SG	12	31	–	✓	✓	✓	✓	–	–
49	McGill	CA	27	41	–	✓	✓	✓	✓	✓	✓
50	CUHK	HK	32	41	–	✓	✓	✓	–	–	✓
51	CMU	US	52	24	–	✓	✓	✓	✓	–	✓
52	Wisconsin	US	–	53	36	✓	✓	–	✓	✓	✓
53	USTC	CN	–	51	40	**P**	✓	✓	✓	–	–
54	WUSTL	US	–	67	26	✓	–	–	✓	–	✓
55	Monash	AU	36	58	–	✓	✓	–	–	–	✓
56	SNU	KR	38	58	–	✓	✓	✓	✓	–	–
57	Heidelberg	DE	–	49	51	✓	✓	✓	✓	–	✓
58	HKUST	HK	44	58	–	✓	✓	✓	✓	–	✓
59	Karolinska	SE	–	53	50	✓	**P**	✓	✓	✓	✓
60	TU Delft	NL	47	57	–	✓	✓	✓	✓	✓	✓
61	ANU	AU	32	73	–	✓	✓	**P**	✓	–	✓
62	KU Leuven	BE	60	46	–	✓	✓	✓	✓	✓	✓
63	Queensland	AU	42	–	65	✓	✓	✓	✓	–	✓

Appendix D. Documentary matrix, Ecuadorian census

Legend: ✓ present; – absent; **P** partial, meaning limited scope, for example postgraduate programmes, admissions or a mobile application only; *NV* not verifiable by this method, meaning the site could not be retrieved, the document was not machine-readable, or the element is rendered dynamically and could not be detected. Type: P public, C co-financed, A self-financed. Columns: Notice, LOPDP (the law is named expressly), Rights, DPO, Acc. (accessibility statement or tools declared), Transp. (LOTAIP transparency section published; this column records transparency under the Ecuadorian access-to-information act and is not the transport-security indicator of Table 1). Institutions created in 2024 or later are marked ‡. Coding date: August 2026. The live cookie measurements are reported in aggregate in Section 4.2 and per site in the deposited dataset; they are not reproduced here, because the documentary and live indicators are different constructs and presenting them in one grid invited the confusion this table avoids.

Table D1. Documentary matrix of the 63 Ecuadorian HEIs.

#	Abbr.	Type	Notice	LOPDP	Rights	DPO	Acc.	Transp.
1	EPN	P	✓	✓	✓	–	–	✓
2	ESPOL	P	✓	✓	✓	✓	–	✓
3	ESPOCH	P	–	–	–	–	–	✓
4	ESPAM MFL	P	–	–	–	–	–	✓
5	ESPE	P	✓	✓	✓	✓	✓	✓
6	UCE	P	*NV*	–	–	–	–	✓
7	UCuenca	P	✓	✓	✓	–	–	✓
8	UG	P	✓	✓	✓	✓	–	✓
9	UNL	P	–	–	–	–	–	✓
10	UAE	P	–	–	–	–	–	✓
11	UTA	P	✓	*NV*	*NV*	*NV*	*NV*	✓
12	UTB	P	✓	✓	✓	✓	–	✓
13	UTC	P	✓	–	–	–	–	*NV*
14	UTMACH	P	✓	✓	✓	✓	–	✓
15	UTM	P	–	–	–	–	–	✓
16	UTN	P	**P**	✓	✓	–	–	✓
17	UTEQ	P	**P**	✓	–	–	–	✓
18	UTELVT	P	–	–	–	–	–	✓
19	UEB	P	✓	*NV*	*NV*	*NV*	✓	✓
20	UNESUM	P	**P**	*NV*	–	–	–	✓
21	UNEMI	P	✓	✓	✓	–	–	✓
22	UEA	P	✓	–	✓	–	–	✓
23	UPSE	P	**P**	✓	✓	✓	**P**	✓
24	ULEAM	P	–	–	–	–	–	✓
25	UNACH	P	✓	✓	✓	✓	✓	✓
26	UNAE	P	–	–	–	–	–	✓
27	UPEC	P	–	**P**	–	–	–	✓
28	Yachay Tech	P	–	–	–	–	–	✓
29	Ikiam	P	✓	✓	✓	✓	–	✓
30	UArtes	P	✓	✓	✓	✓	–	✓
31	Amawtay Wasi	P	–	–	–	–	–	✓
32	USECIPOL‡	P	**P**	–	–	–	–	✓
33	IAEN	P	✓	✓	✓	–	✓	✓
34	UASB	P	✓	✓	✓	✓	–	✓
35	FLACSO	P	–	–	–	–	–	✓
36	PUCE	C	✓	✓	✓	✓	–	✓
37	UCSG	C	✓	✓	✓	✓	–	✓
38	UCACUE	C	✓	✓	✓	✓	✓	✓
39	ULVR	C	–	–	–	–	–	✓
40	UTPL	C	✓	✓	✓	✓	–	*NV*
41	UTE	C	✓	–	–	–	✓	*NV*
42	UDA	C	–	–	–	–	–	✓
43	UPS	C	✓	✓	✓	–	–	✓
44	UISEK	A	✓	–	✓	–	–	–
45	UEES	A	✓	✓	✓	✓	–	–
46	USFQ	A	✓	✓	✓	✓	–	–
47	UDLA	A	✓	✓	✓	✓	–	–
48	UIDE	A	✓	✓	✓	✓	–	✓
49	UNIANDES	A	✓	–	✓	–	–	–
50	UPACIFICO	A	*NV*	–	–	–	–	–
51	UTI	A	✓	–	✓	–	–	✓
52	Casa Grande	A	✓	✓	✓	✓	–	–
53	UTEG	A	✓	✓	✓	✓	–	✓
54	UISRAEL	A	✓	*NV*	*NV*	*NV*	*NV*	*NV*
55	UDET	A	–	–	–	–	–	–
56	UMET	A	✓	*NV*	*NV*	*NV*	–	✓
57	U. Otavalo	A	*NV*	*NV*	–	–	–	*NV*
58	USGP	A	✓	✓	✓	✓	–	✓

Continued on next page

(Table D1, continued)

#	Abbr.	Type	Notice	LOPDP	Rights	DPO	Acc.	Transp.
59	U. Hemisferios	A	✓	✓	✓	✓	–	✓
60	UNIB.E	A	✓	–	✓	–	–	–
61	ECOTEC	A	✓	✓	✓	✓	–	✓
62	UDR	A	–	–	–	–	–	–
63	UBE	A	✓	✓	✓	✓	✓	–

Appendix E. Supplementary figures

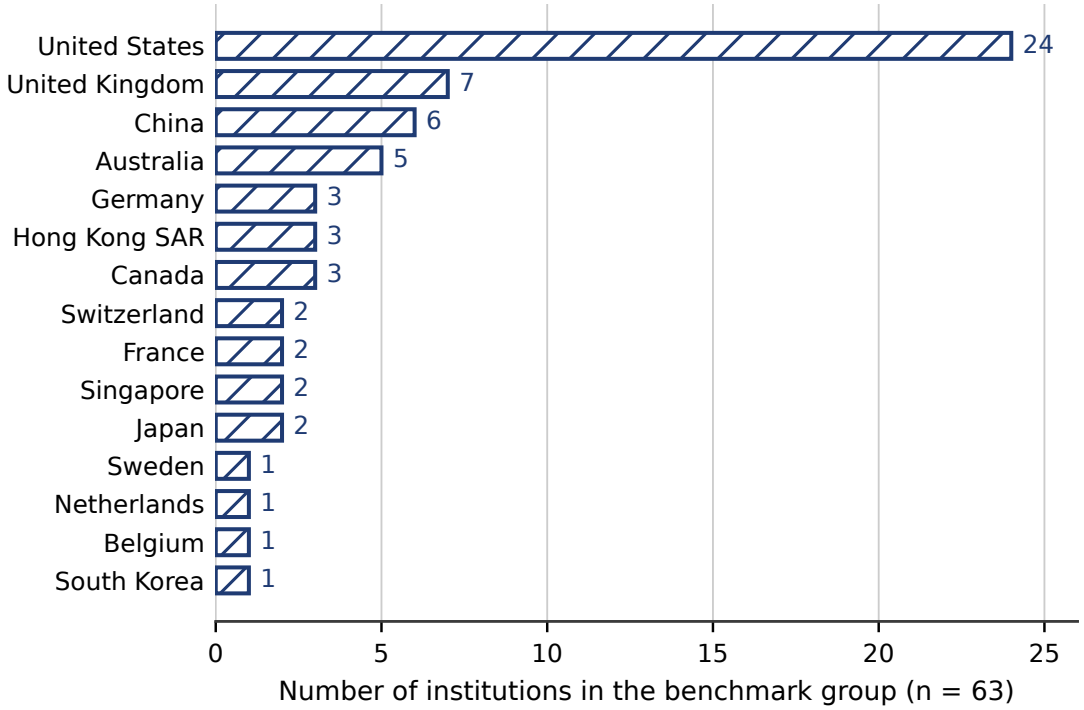


Figure E1. Composition of the benchmark group by country. The group is the extreme upper tail of three international rankings and is not a sample of the world's universities; see Section 3.1.

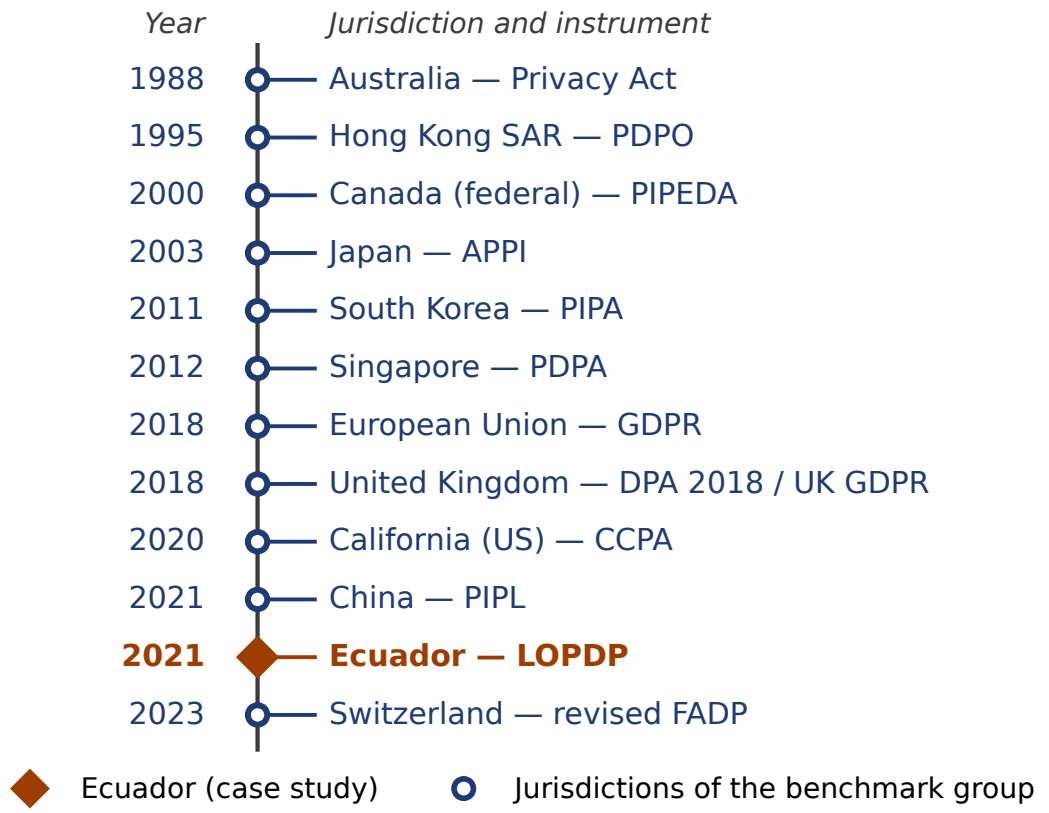


Figure E2. Year in which the current general data-protection regime took effect in each jurisdiction of the study. For the United States, which has no comprehensive federal statute, California's CCPA is used. Ecuador and China, both in 2021, belong to the most recent wave, followed only by Switzerland's revised act in 2023.